THE AUDIBLE FUTURE: DOLBY ATMOS AND THE IMMERSIVE SOUND OF CONTEMPORARY SCIENCE FICTION

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ABSTRACT

This thesis examines the aesthetics and ideology of the recent wave of so-called "immersive" sound formats designed for cinematic exhibition. Focusing on the Dolby Atmos format, this paper contextualizes contemporary audiovisual aesthetics within the cultural, industrial, and technological history of American cinematic sound, and argues that Atmos forecloses possibilities for sonic experimentation by forcing strict adherence to current spatially based conventions for cinematic realism. Formal analysis of recent science fiction films—including *Guardians of the Galaxy*, *Transformers: Age of Extinction*, and *Teenage Mutant Ninja Turtles*—reveals the tension between the purported mimetic naturalism of immersive sound and the irreal Atmos soundtracks of contemporary Hollywood spectacles.

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INTRODUCTION

With the release of Pixar's *Brave* into just 14 Atmos-equipped theaters in summer 2012, Dolby quietly launched its latest cinematic sound format.[1] As the preeminent company creating film sound technology and influencing audiovisual aesthetics, Dolby's introduction of Atmos is a meaningful development for the creation and exhibition of motion pictures. Since the early success of Dolby Stereo as the first widely adopted multi-channel theatrical format in the late 1970s and early 80s, Dolby technology has remained an industry standard for sound both theatrically (with successive formats Dolby SR, Dolby Digital, and Dolby Digital EX) and in the home (as one of two standard audio formats for DVD and Blu-ray). Atmos serves as an indication of the direction of sound technology and aesthetics as Dolby's entry into the growing market of "3D" or "immersive" sound, a loosely defined term referring to new audio formats that add additional speakers—most notably to the ceiling—beyond the now standard 5.1 or 7.1 arrangement.[2] By dint of Dolby's longstanding relationships with studios and exhibitors, Atmos has quickly become the most successful immersive audio format in terms of the number of movie soundtracks produced and number of equipped theaters. Despite besting its competitors,

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[1] Smith, Jeff. 2013. "Atmos, All around: A Guest Post by Jeff Smith." *Observations on Film Art*. May 1. http://www.davidbordwell.net/blog/2013/05/01/atmos-all-around-a-guest-post-by-jeff-smith/.

[2] The terms "3D sound" and "immersive sound" are used interchangeably by Barco's Auro 11.1, which is Atmos's chief competitor for post-5.1 exhibition. Another competitor, IOSONOS markets its technology as 3D sound to match 3D visuals, and highlights spatial representation as the company's primary aim. Dolby primarily uses the term "immersion" to describe Atmos. See "Auro 11.1 - Barco's Immersive Cinema Sound Format." 2015. "Cinema & Entertainment: Solutions for Cinemas." *IOSONO: The Future of Spatial Audio*, and "Dolby Atmos Multidimensional Sound." 2014. *Dolby Atmos*. It is also worth noting that the metaphor of immersion has been consistently used in both marketing and scholarship for previous stereo formats. In order to avoid confusion, I will refer to these new formats as "immersive" or "3D" sound, and previous formats by speaker arrangement, i.e. mono, 5.1, 7.1, etc. I use this terminology primarily out of convenience and to reflect how the technology is being marketed, not as a qualitative judgment on how various formats "immerse" the viewers in sound.

Dolby Atmos—like all immersive audio formats—is still struggling for widespread adoption in film production. Compared to the company's previous formats, Dolby Stereo and Dolby Digital, Atmos has been slow to catch on with filmmakers and exhibitors after nearly three years of aggressive marketing. Still, Atmos is significant as the first notable sound format designed for the era of digital projection in cinema. Freed from the physical and spatial limitations of celluloid film, Dolby has introduced a format that would be technically impossible without the affordances of digital production and the capacious delivery system of the Digital Cinema Package. Regardless of the eventual success or failure of Dolby Atmos itself, the emerging technology and aesthetic practices in the early years of digital projection that will shape the future of cinematic sound through the precedent of non-channel-based algorithmic sound mixing.

Somewhere in the introduction of nearly all works of film sound scholarship, there comes a moment in which the author bemoans the marginalization of sound in the study of film and notes the dearth of quality writing on the subject.[3] Because this observation that film sound remains understudied is so common, many scholars make a move to self-reflexively acknowledge the hackneyed prevalence of such a declaration—I'm making that move right now.[4] The statement that film sound is understudied has become something of a truism, but it is still true. In the past thirty years, scholars have slowly developed a vocabulary and methodology

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[3] Rick Altman cogently made this oft-repeated observation in the introduction to the seminal "Cinema/Sound" issue of *Yale French Studies*, writing "More than a half century after the coming of sound, film criticism and theory still remains resolutely image-bound. Early filmmaker's skepticism about the value of sound has been indirectly perpetuated by generations of critics for whom the cinema is an essentially visual art, sound serving as little more than a superfluous accompaniment." See "Introduction." *Yale French Studies*, no. 60 (January 1, 1980): 3. Scholars have continued to reiterate this point in the subsequent decades. See Michel Chion, *Audio-Vision* (New York: Columbia UP, 1994), xxv; William Whittington, *Sound Design and Science Fiction* (Austin: U of Texas P, 2007), 8; K.J. Donnelly, *Occult Aesthetics* (New York: Oxford UP, 2014), 12.

[4] See Mark Kerins, *Beyond Dolby (Stereo)* (Bloomington: Indiana UP, 2010), 3.

with which to study film sound aesthetics. Yet most formal analysis of film sound remains focused on the text of the film itself, ignoring the specific industrial forces that shape filmmakers' sonic approaches.[5] On the other end of the spectrum, the broader field of "Sound Studies," as demarcated in Jonathan Sterne, tends to focus on material and cultural at the expense of formal analysis of sound itself.[6] My thesis aims to combine these two approaches by showing how our cultural understanding of stereo sound reproduction and the industrial history of sound technology inform the specific. I argue that the cultural history of sound reproduction is inscribed in the film soundtrack, and thus the historical approach of Sound Studies should be applied to the study of film sound aesthetics.

Sound studies, as it is currently constituted, was effectively formulated by Jonathan Sterne effectively founded with the publication of his book *The Audible Past: Cultural Origins of Sound Reproduction* in 2003. Sterne writes a history not of sound technology itself, but the ideological and cultural basis that allowed for the possibility of sound reproduction.[7] Employing an approach heavily indebted to historical materialism,[8] he demonstrates how common-sense understandings of hearing are not based on biological or psychological facts, but on a culturally

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[5] This formal approach to film sound aesthetics is typified by much of the early work on film sound done by Michel Chion, Rick Altman, Elisabeth Weiss and others. Essays that examine sonic aesthetics from an auterist perspective are common and tend to view the soundtrack as a pure expression of creative genius. For examples of this approach, see "Part III: Practice" in Eds. Elisabeth Weis and John Belton, *Film Sound* (New York: Columbia UP, 1985).

[6] This material historical approach is typified by Sterne, who is obviously not the only scholar in the field, but has had an outsized influence on the current shape of sound studies. See Sterne, Jonathan. "Sonic Imaginations." In *The Sound Studies Reader*, edited by Jonathan Sterne, (New York: Routledge, 2012), 1–21.

[7] See Jonathan Sterne, *The Audible Past* (Durham: Duke UP, 2003) 2.

[8] Sterne provides some lucid, if generous, glosses of Horkheimer and Adorno's work on sound recording in the *Dialectic of Enlightenment*. Sterne, *The Audible Past*, 157-8.

constructed theory of the senses informed by Romanticism and theology.[9] Of particular relevance to my project is Sterne's exploration of the cultural construction of sound fidelity as it relates to the mediation of reality, which is a primary concern of all film sound technology. Sterne builds on Walter Benjamin's insights on aura and presence, making use of Benjamin's key insight that the copy creates the very idea of the original, which did not exist before the possibility of sound reproduction.[10] Sterne argues that early sound technology companies created the concept of fidelity through marketing campaigns designed to equate reproduction with the "original event." As a commercial category, "high fidelity" relates more to incremental changes in existing sound technologies than to a closer relation to an "original" sound event.[11]

Questions of mediation and fidelity are complicated when the artificially constructed, fantastical soundtracks of science fiction film. Historically, science fiction motion pictures have had innovative soundtracks, and the genre helpfully foregrounds its relationship to technology in its thematic and narrative concerns.[12] As previously mentioned, the successful introduction of Dolby Stereo surround sound was closely related to the science fiction genre with the success of *Star Wars*. Because of the speculative nature of science fiction, sound designers are able to experiment more with their approaches to constructing the soundtrack. There is no correct way to realistically represent something inherently fantastical like alien life or intergalactic space travel, so filmmakers have free reign to experiment with new approaches, as long as they don't stray too far from standard conventions of cinematic realism—specifically the spatial indicators provided

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[9] In particular, what Sterne dubs "the audiovisual litany" calls into question many of the assumptions that inform seminal sound scholarship in film studies. Sterne, *The Audible Past*, 14-6.
[10] See Walter Benjamin, "The Work of Art in the Age of Mechanical Reproduction," Trans. Hannah Ardent. *Illuminations*, (New York: Shocken), 217-51.
[11] Sterne, *The Audible Past*, 261-74.
[12] William Whittington, *Sound Design and Science Fiction*, 3-4.

by contemporary soundtrack aesthetics. Two of the most celebrated sound designers working in film, Ben Burtt and Walter Murch, were first celebrated for their innovative sound design in science fiction films in *Star Wars* and *THX 1138* (1971) respectively. Aesthetic appreciations of film sound refer often to a canon of innovative science fiction soundtracks such as *Blade Runner* (1982), *2001: A Space Odyssey* (1968), and first two entries in the *Alien* franchise (1979 and 1986).[13] As such, the sound of science fiction films has historically provided a rich text for emerging and innovative approaches to the soundtrack.

In Chapter 1, I explore the ideology of realism represented by Dolby Atmos and immersive sound contextualized in the cultural and technological history of cinematic sound. I argue that compared to previous formats, Atmos forecloses sonic experimentation by forcing strict adherence to current spatially based conventions for cinematic realism. In Chapter 2, I examine *Guardians of the Galaxy*. As the second highest-grossing movie of 2014 in the United States is exemplary of Hollywood filmmaking that is both critically and commercially succesful.[14] I compare the movie's audiovisual aesthetics to director James Gunn's previous films, sound designer David Acord's other sound work, and *Star Wars*, a landmark Dolby Stereo film and clear inspiration for *Guardians*, narratively and aesthetically. Additionally, I analyze the sonic aesthetics of the four movies released with Atmos soundtracks on Blu-ray as of December 2014: *Transformers: Age of Extinction*, *Teenage Mutant Ninja Turtles*, *The Expendables 3*, and

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[13] Whittington touches on each of these films at some length in his monograph *Sound Design and Science Fiction*.

[14] "2014 Domestic Grosses." 2015. *Box Office Mojo*. Unfortunately, I did not have an opportunity to see the year's highest-grossing movie in the United States, *The Hunger Games: Mocking Jay – Part 1*, in Dolby Atmos. The format is still somewhat difficult to study due to the small number of equipped theaters in any one metropolitan area—there is only one Atmos screen near Washington, DC. The slow trickle of Atmos Blu-ray discs means that, for now, the cinema is the only place to hear the format.

Step Up All In.[15] Dolby Atmos represents the latest development in film sound technology's long history of marketing and selling new technology as mimetically accurate. This history of cinematic sound is a history of the ideological confusion surrounding the possibility of mediating reality—and Dolby is perhaps the single most influential company to shape common understandings and approaches to cinematic realism. The specific approach to realism promised by the Atmos format makes sense more as a refinement of existing 5.1 surround technology than as an aural recreation of the auditor's lived experience. Atmos codifies what Mark Kerins calls the prevalent "Digital Surround Style" of 5.1 surround sound, and offers surprisingly little affordances for stylistic applications that do not subscribe to this aesthetic strategy that primarily foregrounds spatial relationships over other sonic detail. To fully understand Atmos, we have to understand its place in the history of film sound. As the subsequent historical overview of exhibition formats shows, like Atmos, each new iteration of film sound technology has been marketed to theater owners and moviegoers as a more accurate representation of reality.

[15] As of February 2015, Dolby has announced or released only five additional titles: *Gravity*, *The Hunger Games: Mocking Jay – Part 1*, *John Wick*, *On Any Sunday: The Next Chapter*, and *Unbroken*. This is small fraction of the 150 movies mixed in Atmos—55 of which were released in 2014. Disney Studios, which ultimately owns *Guardians of the Galaxy*, has not yet announced any Blu-ray titles to be released with Atmos soundtracks. See "Dolby Atmos Movies on Blu-Ray." 2015. Accessed February 16. /us/en/experience/dolby-atmos/bluray-and-streaming.html.

CHAPTER 1: UNDERSTANDING ATMOS: DECODING IMMERSIVE SOUND

Creating the Sound of Cinema: The Mono Era

The history of film sound from the silent era to the present day illustrates how sound aesthetics and technology are closely tied to the economic interests of the film industry. The transition in America and Europe from silent to sync sound cinema occurred rapidly in the late 1920s and early 1930s thanks to Hollywood's aggressive expansion of the profitable "talkies" into new markets in the United States and abroad.[16] During this transitional era, filmmakers and sound technicians wrote extensively about the possibilities for emerging film sound aesthetics, but after the codification of standard sound practices by the Hollywood studio system in the early 1930s, sound ceased to be a major concern in most writing on film. [17] Early sound films treated synchronized dialogue and sound effects as a novelty, only to be employed for moments of spectacle or heightened emotion as in the musical performances in *The Jazz Singer*. Heady theorization on the possibilities of contrapuntal sound and the tyranny of dialogue ruining the art of cinema quickly disappeared as studios turned sound into a below-the-line technical (as opposed to artistic or creative) craft. By the mid 1930s, US film studios had developed large sound departments run by a Sound Director or Director of Sound who primarily acted as an administrator overseeing a large staff.[18] Any one member of the sound department specializing in one aspect of new film sound technology might be working on individual reels of myriad films at

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[16] See Gomery, Douglas, "The Coming of Sound," *Film Sound*, Eds. Elisabeth Weis and John Belton, (New York: Columbia UP, 1985), 5-25; Douglas Gomery, "Economic Struggle and Hollywood Imperialism," *Film Sound*, 25-36.

[17] For an overview of early writing on film sound see the "Classical Sound Theory" section in Eds. Elisabeth Weis and John Belton, *Film Sound*, 75-144.

[18] Cousins, Mark. "Walter Murch." In *Projections: Film-Makers on Film-Making*, edited by John Boorman and Walter Donohue, (London: Faber, 1996), 151-2.

any one time. Individual studios developed house styles based on their genre specialties and sound libraries, but it was relatively rare for sound films in the 1930s and 40s to have unique soundtracks—films would feature the same library of soundtracks and share the same general aesthetic strategies.[19] While some star cinematographers like Gregg Toland and Charles Lang were celebrated for their formal innovations, there are few if any filmmakers celebrated primarily for their aural style in the first decades of sound film.

Beyond the studio's organizational structure relegating sound to a technical afterthought, the physical affordance of the filmstrip itself limited the representational capabilities of film sound. After early experiments with cumbersome sound-on-disc formats, the film industry settled on the monophonic optical sound-on-film as the standard for cinematic exhibition. The optical soundtrack is a format that has never had any significant applications outside of film exhibition, being created as the cheapest, most convenient solution for syncing image and sound. The RCA Photophone system established the template for the optical soundtrack in which a variable area waveform was printed on the filmstrip next to the image track. During projection, a separate low-wattage bulb shines light through the printed optical soundtrack, which is converted by a photoelectric cell into electric impulses, and ultimately sound.[20] As a format born of convenience, optical sound has a small dynamic and frequency range, meaning that a limited amount of audio information can be included on the soundtrack before masking—the effect of one sound source rendering another inaudible—occurs. The optical soundtrack fades slightly with each projection, making the soundtrack increasingly noisy through the life of a print.

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[19] For more on the "house style" of Hollywood studios see Whittington, William. *Sound Design and Science Fiction*, (Austin, TX: U of Texas P, 2007), 101, 112.
[20] Handzo, Stephen. "A Narrative Glossary of Film Sound Technology." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, (New York: Columbia UP, 1985), 387-8.

Furthermore, exhibitors were reluctant to upgrade the primitive sound systems purchased in the early rush to capitalize on the talkie craze. So even as Hollywood sound departments made use of increasingly sophisticated microphones and sound editing machines, the quality of sound in the theaters remained stagnant. The intransigence of exhibitors to upgrade for sound led to the Academy of Motion Picture Arts and Sciences adopting the "Academy Curve" as the standard theater playback response curve (the dynamic and frequency range of speakers) in 1938.[21] The Academy Curve, based on the limited capabilities of the optical soundtrack and early speaker systems, remained the industry standard until Dolby's intervention in the mid 1970s.

It is clear that early film sound technology and aesthetics were not developed according to a Bazin-like drive for greater realism—they were created according to the industrial practices and technological affordances of the time. During the 1930s, sound departments developed conventions for cinematic realism that shaped subsequent developments and approaches to film sound. First, the limited frequency response of the Academy Curve meant that the soundtrack was best-suited for reproducing the human voice, which contributed to the development of a verbocentric cinema in stark contrast to the vaunted ideal of universal visual language in late silent film.[22] Thanks in part to the affordance of the Academy Curve, the clear, audible mixing of dialogue remains a cardinal rule for film sound to this day, with very few exceptions.[23] Second,

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[21] Gianluca Sergi, *The Dolby Era*, (New York: Manchester UP, 2004), 14-5.

[22] Michel Chion coined the term *verbocentric* to refer to classical Hollywood sound conventions in which speech is the primary focus of the viewer's attention and is complemented by visual action. See Chion, Michel. *Film, A Sound Art*. Translated by Claudia Gorbman, (New York: Columbia UP, 2009), 73. Chion goes into great detail about the emergence of early sound conventions in Chapters 3 and 5 of that volume.

[23] Director Christopher Nolan's claims of formal experimentation aside, the occasional masking of inconsequential dialogue in his 2014 film *Interstellar* by music and sound effects was widely decried as a failure of sound design in many media outlets. See Giardina, Carolyn. "Christopher Nolan Breaks Silence on 'Interstellar' Sound." *The Hollywood Reporter*, November 5, 2014.

due to the aforementioned problem of masking and the usual placement of the single speaker behind the screen, early sound conventions tended to eschew extraneous offscreen sounds. Sound departments primarily limited the soundtrack to representing only what was shown onscreen, leading to the persistent disparaging description of sound aesthetics as "see a dog, hear a dog."[24] With sounds primarily accompanying onscreen actions and emanating from behind the screen, evolving audiovisual aesthetics created a strict spatial relationship between sound and image. In Hollywood films, the vast majority of shots were constructed so that point of audition and point of view were effectively the same. Because of the difficulties of sound editing, early talkies necessarily made use of direct sound, making the spatial relationship between sound and image less an artistic choice than a technical problem to be solved. In the 1930s, Western Electric executive J.P. Maxfield published two enormously influential essays in industry trade journals arguing for a consistent spatial relationship between sound and image ostensibly based on scientific observations on human phenomenology, but clearly limited by the possibilities of early sound technology.[25] Following Maxfield's cue, soundtracks were created as if the auditor is listening from the same position with which the camera views the scene, with a directional microphone pointed the same direction as the camera cutting out any sounds not emanating directly in front of the listener.[26] The famous chase scene in John Ford's seminal 1939

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http://www.hollywoodreporter.com/behind-screen/christopher-nolan-breaks-silence-interstellar-749465. Robert Altman is one of the few major Hollywood directors to flout the rule of always audible dialogue, most notably in some early scenes in *McCabe & Mrs. Miller* (1971).

[24] Kenny, Tom. "Introduction." In *Sound for Picture: An Inside Look at Audio Production for Film and Television*, edited by Jeff Forlenza and Terri Stone, (Emeryville, CA: Hal Leonard Corp, 1993), v-vi.

[25] Altman, Rick. "Sound Space." In *Sound Theory, Sound Practice*, 46–64.

[26] Dialogue is the one notable exception to the convention of matching point of audition and point of view. Regardless of camera placement and the point of audition implied by the soundscape, dialogue is almost always clearly audible.

Stagecoach typifies this practice. The sound of hooves matches the objective view of the camera as Ford cuts from shots of the interior of the stagecoach, to the exterior, to shots of Apaches on horseback. The sounds of hooves and the rattle of the stagecoach are not at a consistent volume through the scene, nor does the soundtrack match one character's point of view. The soundscape slightly changes with each cut. It may seem obvious now to note the spatial tie between the image and soundtrack, but this particular set of sonic conventions was not a forgone conclusion in the early years of sound cinema as prospective essays on the possibilities of contrapuntal and asynchronous sound attest.[27]

Cinematic sound was developed and codified at a specific historical moment based on particular economic, industrial, and technological formations. Writing on audio formats and format theory, Jonathan Sterne notes that media formations must be understood as "particular historical instantiations, and not peaks towards which progress has been climbing."[28] Accordingly, later film sound formats must be understood as products of their particular cultural context, but also in relationship to preceding sound technology and aesthetic conventions. All subsequent cinematic audio technologies were created in relation to the automatisms (to borrow Stanley Cavell's terminology) codified by the Hollywood studio system during the early mono era of sound film.

Listening to Widescreen: Magnetic and Stereophonic Sound

Through the 1930s and 1940s, cinematic exhibition saw no major technical changes, with the vast majority of films shown at the 1.37:1 Academy ratio and heard in monophonic sound

[27] See Eisenstein, Sergei, Vsevolod Pudovkin, and G.V. Alexandrov. "A Statement," translated by Jay Leyda, 83–85, and Pudovkin, Vsevolod. "Asynchronism as a Principle of Sound Film," translated by Marie Seton and Ivor Montagu, 86–91. Both in *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, (New York: Columbia UP, 1985).
[28] Sterne, Jonathan. *MP3: The Meaning of a Format*, (Durham: Duke UP, 2012), 11-2.

within the frequency range of the Academy Curve.[29] After World War II, the popularity of television was significantly cutting into ticket sales, threatening the profitability of theatrical exhibition. In response to this threat, film studios sought to distinguish the cinematic experience from the televisual medium. This push for differentiation manifested in a rash of widescreen formats, elongating the near-square of the Academy ratio to the now-common rectangular widescreen aspect ratios.[30] In addition to a larger image, the new widescreen formats also introduced the first significant adoption of stereophonic sound for cinematic exhibition, with the different formats requiring between four and seven speakers. Most widescreen formats included multiple audio tracks for three reasons. First, stereo sound was an obvious way to distinguish film from television, which was an exclusively monophonic medium into the 1980s.[31] Second, the wider 70mm filmstrip used in formats like CinemaScope and Todd-AO simply had more space to work with, making it possible for films to include a greater amount of audio information.[32] Finally, studios felt it necessary to add additional speakers to match the wider screens in accordance with the conventions of realism developed in the mono era that created a

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[29] In 1940, Warner Brothers introduced "Vitasound," a three-channel stereo format with one speaker behind the screen dedicated to dialogue, and two surround speakers playing music and sound effects. Later that year, Disney introduced "Fantasound" for the release of *Fantasia* (1940). This format featured four audio tracks, three of which were sent to three dedicated speakers in the front of the auditorium, and one of which was sent to an array of speakers around the audiotirum. Fantasound required a second separate filmstrip for its optical soundtrack. The prohibitive expense of adding additional speakers and a second projector devoted to sound meant the system was only installed in six theaters, and only used for screenings of *Fantasia*. Although Vitasound and Fantasound were never widely adopted, their approach to stereo sound was later modified for various widescreen formats. See Kellogg, Edward W. "Final Installment: History of Sound Motion Pictures." *SMPTE Journal* 64, no. 8 (1955): 429–30.

[30] Widescreen aspect ratios ranged between1.66:1 and 2.77:1 based on the technical vagaries of the various formats. For more on the industrial and technological forces that shaped the competing formats see Belton, John. *Widescreen Cinema*, (Cambridge, MA: Harvard UP, 1992).

[31] Hull, Joseph. "Surround Sound Past, Present, and Future." Dolby Laboratories, 1999.

[32] Belton, *Widescreen Cinema*, 152.

strong spatial connection between sound and image. Because sound closely matched onscreen action, most speakers were added behind the screens in order to ensure that sounds would emanate from roughly the same location as their onscreen source.[33] In the name of realism, the speaker placement of widescreen format emphasized movement onscreen, unlike later Dolby formats that emphasized the immersive possibilities of surround speakers placed around the audience.

Early CinemaScope films such as *The Robe* and *How to Marry a Millionaire* employ extreme stereo separation for onscreen dialogue and sound effects to closely match the spatial position of the sound to their apparent onscreen sources.[34] This close spatial matching led to the short-lived mixing practice of "traveling dialogue," where the sound of a voice would pan across the front speaker array following a character's onscreen movements. Although this practice ostensibly follows spatially-based Hollywood aesthetic conventions, critics (most notably Bosley Crowther in the *New York* Times) found the panning of dialogue to be distracting, and traveling dialogue gave way to the still-common practice of restricting dialogue to the front center channel regardless of the source's onscreen (or offscreen) location within the diegesis.[35] Even as mixers stopped panning dialogue, ambient and spot sound effects continued their strict spatial

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[33] Ibid, 151.

[34] My observations of CinemaScope aesthetics are based on the sound mixes on the DVD releases for *The Robe* and *How to Marry a Millionaire*, which are the first two films released in Cinemascope. The DVD releases for both movies incredibly include the original 4.0 mix, which is a somewhat unusual act preservation for a major studio home release where upmixing the original mix to 5.1 is standard practice. In a standard 5.1 home theater setup, the front left, center, and right channels match the original speaker arrangement, and the two surround channels reproduce the track sent to the single rear speaker installed in CinemaScope theaters.

[35] For more on the controversy over traveling sound see Belton, John. "1950s Magnetic Sound: A Frozen Revolution." In *Sound Theory, Sound Practice*, edited by Rick Altman, 161-4, and Crowther, Bosley. "Sound and (or) Fury: Stereophonic System Is Debated in Hollywood." *The New York Times*, January 31, 1954.

correspondence with their visual counterparts. The disappearance of traveling dialogue with the marriage of dialogue to the front center speaker is the most conspicuous indication of the often-arbitrary relationship between audiovisual aesthetics and an objectively accurate representation of reality. Through the mediation of *more* auditory and visual information, widescreen formats supposedly promised greater mimetic accuracy according to a logic of representation in which more information (i.e., more screen, more sound) is equated with greater realism. But most filmmaking, like most sound recording, is what Jonathan Sterne calls a 'studio art,' meaning that there is no original event that the medium reproduces.[36] Movies are created for and through mediation—through the idiosyncratic process of editing, the artificial synchronization between sound and image, and the addition of non-diegetic material—so attempting to define realism according to the final work's indexical relationship to a nonexistent original event is misguided.[37] With no stable referent to mediate, it follows that Hollywood realism conventions that spurred the implementation of stereo sound and traveling dialogue would prove to be estranging to audiences. Despite the purported technological advancements of widescreen formats, audiences grew to associate the systems with spectacle rather than realism.[38]

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[36] Sterne, Jonathan. *The Audible Past: Cultural Origins of Sound Reproduction*, (Durham: Duke UP, 2003), 225-40.

[37] Writings on the indexical nature of film tend to focus exclusively on the image track, ignoring the artificial construction of the soundtrack and the interplay between sound and image. See Altman, Rick. "Introduction: Four and a Half Film Fallacies." In *Sound Theory, Sound Practice*, 42-5.

[38] Belton, John. "1950s Magnetic Sound: A Frozen Revolution," 155-6.

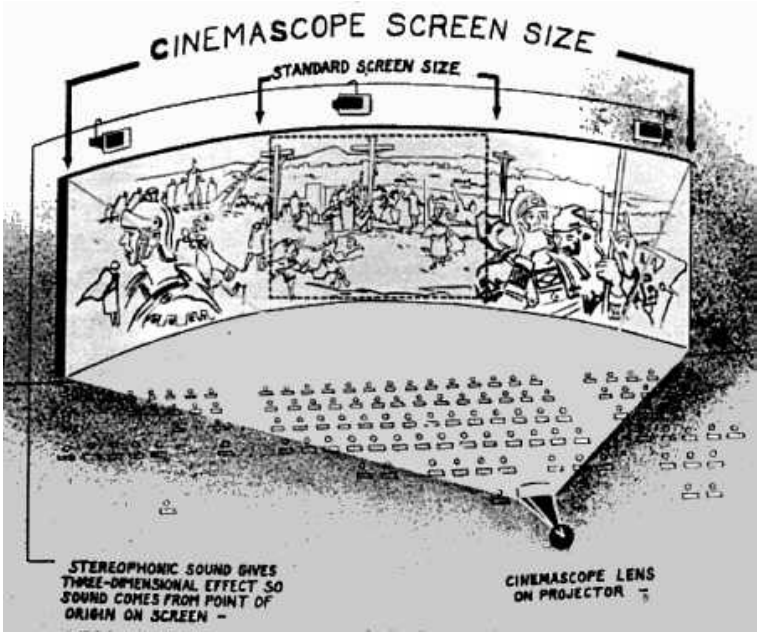

Figure 1. This promotional illustration highlights the spatial rationale for the three speakers placed behind the screen in Cinemascope. It fails to represent the fourth speaker, which would be placed behind the audience. "Cinemascope: What It Is: How It Works." *American Cinematographer*, March 1953.

Ultimately, studios succeeded in persuading exhibitors to upgrade theater screens to accommodate wider aspect ratios, but failed to convince the majority of theater owners to install stereo sound systems, largely due to the additional costs they incurred. Many widescreen formats, on both 70mm and 35mm, dispensed with optical sound in favor of magnetic soundtracks, which offered expanded frequency and dynamic ranges, and a lower signal-to-noise ratio.[39] Yet the expense of additional speakers and new equipment to play magnetic soundtracks proved to be prohibitive to theaters projecting 35mm prints. Most exhibitors continued showing films in mono using optical sound, and eventually the production of stereo soundtracks slowed to a trickle.[40] Stereo magnetic soundtracks survived only on 70mm prints used in formats like

[39] Ibid, 155.
[40] In addition to exhibitor ambivalence, the cost of producing filmstrips with magnetic soundtracks led to the reversion to optical sound. Optical soundtracks were printed along with the image track, but adding a magnetic soundtrack required another step and additional expense in the production of each filmstrip. Ibid, 157.

Todd-AO, and became increasingly associated with the spectacle of musicals and epics. The success of 1950s stereo sound was limited, but widescreen formats had some important implications for the future technological development of cinema sound. Most importantly, widescreen cinema helped crystalize the belief that adding speakers and channels was the primary way to improve cinema sound. Additionally, the two-tiered exhibition environment of competing mono and stereo sound created a tension between realism and spectacle in film sound, with stereo becoming associated more with spectacle and genre films. Finally, the long term success of certain widescreen formats, particularly Todd-AO, was influential on Dolby's subsequent aesthetic and technological approaches that set the standard for the sound of film

The Dolby Revolution: From Stereo to Digital

By the end of the 1950s, only one-quarter of all theaters had converted to stereo, leading studios to largely give up on creating stereo mixes for 35mm prints.[41] Through the 1960s, several hundred theaters continued to screen 70mm prints, but the majority of film was heard using the same monophonic optical systems introduced in the early 1930s.[42] Filmmakers, meanwhile, had switched to using magnetic tape throughout production up until the mono optical soundtrack was printed onto the release print. So as sound departments adopted increasingly sophisticated, high fidelity technology to create soundtracks, intransigent exhibitors ensured that film sound aesthetics remained generally stagnant due to the affordances of monophonic sound and the Academy curve.[43] This technical disparity between sound production and exhibition, coupled

[41] Ibid, 156-7.
[42] Belton, *Widescreen Cinema*, 178-9.
[43] Sound aesthetics were not wholly stagnant before Dolby entered the film industry. The dissolution of the Hollywood studio system allowed for a number of "New Hollywood" filmmakers (heavily indebted to the sound experiments of the French New Wave) to begin some remarkable sonic experimentation. But these modernist interventions did not fundamentally alter

with the emerging hi-fi audiophile culture (centered around popular music recording), set the stage for Dolby's transformational entrance into the film industry.[44]

Dolby Laboratories made its name with noise reduction technology designed for both professional recording in the music industry and home stereo use. In the late 1960s, Dolby began its campaign to convince the film industry to adopt its noise reduction technology in exhibition, with an eye towards the lucrative licensing potential of cinema sound.[45] Building off of the recent 1970s fad of quadrophonic (and, briefly, quintaphonic) sound, which used "matrixing" technology to encode four channels of audio into the space of two channels, Dolby created a stereophonic optical sound format for 35mm film.[46] Adapting the speaker arrangement used by Todd-AO, Dolby's new format, called *Dolby Stereo*, made use of six speakers placed around the auditorium: three behind the screen, two in the rear of the auditorium and one subwoofer or low frequency effect (LFE) speaker usually placed in the front of the theater. Sound was decoded from the optical soundtrack by a Dolby processor, which directed the audio information to the appropriate channel.[47]

Thanks largely to the success of blockbuster spectacles like *Superman* (1978) and Star *Wars* (1977), which was one of the first movies to be designed for the new format from before

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predominate conventions of realism. See Jay Beck, "A Quiet Revolution: Changes in American Film Sound Practices, 1967-1979," (Ph.D. diss., University of Iowa, 2003); William Whittington, *Sound Design in Science Fiction*, 55-61.

[44] For further discussion on the influence of popular music hi-fi culture on film sound, see Whittington, *Sound Design & Science Fiction*, 34-7.

[45] Sergi, *The Dolby Era*, 18-23. Throughout this paper, I use "Dolby" to refer to Dolby Laboratories as a corporate entity, not the company's founder, Ray Dolby.

[46] Kerins, *Beyond Dolby (Stereo)*, 30-1.

[47] Dolby continues to use variations of this matrixing technique in contemporary home theater technology. Dolby Pro Logic (in its myriad incarnations) takes 2.0 stereo audio information and, through a proprietary algorithm, automatically creates a 5.1 or 7.1 mix to make use of a home theater's entire speaker array.

production, Dolby Labs succeeded in replacing standard monophonic sound with its proprietary technology.[48] Dolby's success can be partially attributed to their fortuitous partnership with George Lucas, but it was also the result of the careful attention the company paid to the economics of film production and exhibition. Dolby leadership learned from the failure of early stereo formats and drew on the company's success marketing noise reduction technology to audio producers and consumers. Dolby crucially ensured that upgrading to surround sound was relatively inexpensive compared to earlier surround sound formats and guaranteed that Dolby technology was backward compatible, so cinemas could continue to exhibit monophonic prints in addition to Dolby Stereo.[49] The company also sent so-called "Dolby Consultants" to work with the sound team on motion pictures free of charge. These consultants theoretically served as technical guides, showing sound editors and designers accustomed to working in mono how to take advantage of the new surround format.[50] Although these consultants served in an ostensibly advisory role, they effectively assured that Dolby could actively shape surround sound aesthetics according to how the company wished the technology to be used. Due to the technical limitations of Dolby matrixing technology to reliably send a specific sound effect to the correct speaker and the company's aversion to attention-grabbing, "gimmicky" surround effects, Dolby consultants pushed for the technology to be used primarily for unobtrusive ambient sounds.[51]

As the dominant film sound company with employees overseeing the emerging sound aesthetics in the mixing room, Dolby Labs was able to shape how their technology was used in

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[48] Sergi, *The Dolby Era*, 13-26.
[49] Ibid, 17-8.
[50] Sound designer Steve Katz served as the official Dolby Consultant on many key Dolby Stereo releases, including *Star Wars*, *Close Encounters of the Third Kind* (1977), and *Days of Heaven* (1978). See Sergi, *The Dolby Era*, 101-2.
[51] Kerins, *Beyond Dolby (Stereo)*, 34-5.

service of creating an unobtrusively realistic soundtrack in line with standard Hollywood sound conventions based on a strict spatial suturing between sound and image codified in the mono era. Because sounds could not reliably be spatially placed in the surround speakers, Dolby Stereo films used the surround speakers for ambient effects and (non-diegetic) music. The screen remained the primary spatial reference point, but the additional surround speakers could now better reinforce sonic and spatial continuity between disparate shots in a sequence. In the late 1980s, Michel Chion was among the first to note this emerging Dolby Stereo sound aesthetic.[52] Writing at a time when most Hollywood movies were being made in Dolby Stereo, Chion identified the space constructed by the audiovisual realism conventions as the "superfield." He describes the "superfield" as:

> the space created, in multitrack [i.e. stereo] films, by ambient natural sounds, city noises, music, and all sorts of rustlings that surround the visual space and that can issue from loudspeakers outside the physical boundaries of the screen. By virtue of its acoustical precision and relative stability this ensemble of sounds has taken a kind of quasi-autonomous existence with relation to the visual field, in that it does not depend moment by moment on what we see onscreen.[53]

Chion's superfield refers only to the space created by the relationship between the screen and the two surround speakers. The sound emanating from the three speakers behind the screen continued to follow the conventions of close spatial matching from shot to shot developed during the mono era and carried through to widescreen cinema.[54] While the soundscape of the front

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[52] Chion's observations on the spatial aspects of stereo sound were first published in 1988 as "Revolution douce" in *La toile trouée*. (Paris: Cahiers du Cinema/Editions de l'Etoile). The article was published in English three years later. See Chion, Michel. "Quiet Revolution... And Rigid Stagnation." Translated by Ben Brewster. *October* 58 (October 1, 1991): 69–80.

[53] Chion, Michel. *Audio-Vision: Sound on Screen*. Edited and translated by Claudia Gorbman. (New York: Columbia UP, 1994), 150.

[54] Traveling dialogue did not make a comeback with Dolby Stereo. Dialogue remained confined to the front center speaker.

speakers changes from shot to shot to reflect what is onscreen, surround speakers serve to create a non-directional spatial continuity between shots within a scene. Although directionally specific sounds sometimes panned around the entire theater (think of the helicopters in *Apocalypse Now* or the ships and gunfire in *Star Wars*), the surrounds were mainly used to convey a specific sense of place through carefully constructed room tone or recognizable sounds with no identifiable directional source. Dolby executives point to the ambient continuity provided by the sound of softly rustling grass in *Days of Heaven* as an early exemplary use of Dolby Stereo to construct spatial realism.[55] *Blade Runner* provides another notable example of Chion's superfield, with a dense, mostly acousmatic soundscape conveying a hectic, electronic dystopia existing somewhere outside the frame regardless of the onscreen action.[56]

Dolby Stereo remained the cinematic standard until the introduction of digital surround sound in the early 1990s with Dolby's digital SR-D format (eventually known simply as "Dolby Digital") competing with the Sony Dynamic Digital Sound (SDDS) and Digital Theater Systems (DTS) digital formats. The differences between the three technologies are negligible—DTS and SDDS were adopted because of the marketing muscle of Universal Pictures and Sony respectively, not thanks to a noticeably unique aesthetic experience.[57] Digital surround sound allowed for sound designers to make greater use of the surround speakers by solving some of the

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[55] Sergi, *The Dolby Era*, 102. Dolby singles out *Days of Heaven* as important to the eventual success of Dolby Stereo. Terrence Malick was the first director to use the format for a relatively small drama, which weakened its association exclusively with science fiction and action spectacles.

[56] For more on the sound design of *Blade Runner* in its various incarnations see Whittington, *Sound Design and Science Fiction*, 169-88.

[57] Dolby Digital and DTS employ the same 5.1 speaker setup as Dolby Stereo. SDDS adds two additional speakers behind the screen for a 7.1 configuration. These two additional speakers do not significantly alter film sound aesthetics. Most films continue to be mixed for 5.1 sound. Kerins, *Beyond Dolby (Stereo)*, 39-40.

matrixing problems of Dolby Stereo. Sounds could be precisely placed in the surround speakers without having to contend with the frequent errors of matrixed optical soundtracks. Additionally, digital sound afforded greater frequency and dynamic ranges, meaning sound designers had more options to experiment with pitch and volume.[58] As with Dolby Stereo, the new digital formats were designed so that 35mm film prints would be backward compatible for exhibitors who didn't wish to upgrade their sound systems. On filmstrips, the stereo optical soundtrack remained in its place immediately to the right of the image track while the information for the three competing digital formats was squeezed into the remaining space around the sprocket holes.[59] With four separate formats—Dolby Stereo, Dolby Digital, SDDS, and DTS—coexisting on the filmstrip, no physical space remained on 35mm film for additional audio information. It would take the introduction of a new medium for distribution, namely the Digital Cinema Package, for new audio formats to emerge to compete with these three early digital systems.

The switch from analog to digital sound provided filmmakers with new tools for creating soundtracks, but did not seem to provoke an ontological crisis in how audiences understand film sound, unlike the current transition from analog film to digital video that is potentially reshaping our understanding of the moving image.[60] Presaging recent debates about digital images, Eric W. Rothenbuhler and John Durham Peters have argued that analog recording technologies have an authentic relationship to the original because of the causal relationship between mediated sound and its source, whereas digital recording is inauthentic in its translation of sound into semantic

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[58] Ibid, 41.
[59] The digital information for Dolby Digital and SDDS soundtracks is stored on the filmstrip itself. The DTS soundtrack is played off of a separate CD-ROM that syncs with the image via a small timecode on the filmstrip itself. Ibid, 41-3.
[60] See D.N. Rodowick, *The Virtual Life of Film*. (Cambridge: Harvard UP, 2007) and John Belton, "Digital Cinema: A False Revolution."

information, i.e. binary.[61] However, I maintain that from the early days of sound cinema, film soundtracks have had a decidedly inauthentic relationship between reproduced sound and its ostensible source. From the poorly synced ADR lines to the fundamental convention of non-diegetic music, the film soundtrack is a constructed, inauthentic text that follows conventions of realism without having a privileged relationship to reality. As such, digital sound is not appreciably more unreal or inauthentic than analog when it comes to film sound aesthetics.[62] These new digital sound technologies were marketed as being a truer mediation of reality, although their technical and aesthetic improvements were more closely related to previous sound technology than to unmediated human perception. Dolby SR-D follows the commercial logic of iterative improvement by correcting the technical limitations of the analog Dolby Stereo format, but does not really come appreciably closer to replicating how an auditor experiences sound in the real world.

In conjunction with theatrical exhibition, it is also important to bear in mind that 5.1 surround sound was standardized as the DVD audio format, so stereo movies would theoretically sound the same in the theater and at home for the first time.[63] As Barbara Klinger has noted, the home video market is crucial to the economic viability of Hollywood films, so digital surround sound did not become fully entrenched in studio production until it was officially adopted as the new home video sound standard for DVD.[64] Although it's unlikely that the majority of home viewers will ever watch a given film in full 5.1 surround sound, filmmakers can still choose to

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[61] Rothenbuhler, Eric W., and John Durham Peters. 1997. "Defining Phonography: An Experiment in Theory." *The Musical Quarterly* 81 (2): 242–64.
[62] Here I am further indebted to Jonathan Sterne's discussion of the concept of the "original" in sound recording. See Sterne, *The Audible Past*, 219.
[63] Hull, Joseph. "Surround Sound Past, Present, and Future."
[64] Barbara Klinger, *Beyond the Multiplex: Cinema, New Technologies and the Home*. (Berkley: U of California P, 2006), 4-8.

forge an audiovisual aesthetic for this imagined ideal viewer. 5.1 surround sound is the aspirational norm of home video presentation. The possibility of hearing a movie in full surround sound through its post-cinematic afterlife affords directors and sound designers to experiment with the soundtrack, even if many viewers will still encounter the movie in two-channel stereo.

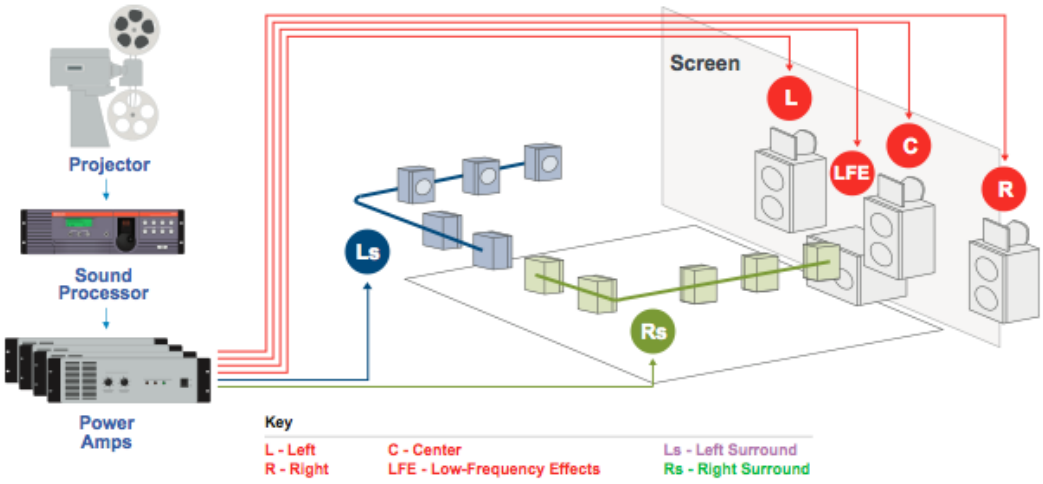

Figure 2. The speaker arrangement remained the same from Dolby Stereo to Dolby Digital. For home theater systems the two surround channels are sent to only two speakers (left surround and right surround), whereas in the theater the two surround channels are sent to an array of surround speakers. "Dolby Atmos Next-Generation Audio for Cinema." Dolby Laboratories, 2014.

Building on Chion's observations on the spatial realist aesthetics of Dolby Stereo, Mark Kerins convincingly argues in *Beyond Dolby (Stereo)* that, because of the affordances of digital surround sound and its standardization in cinemas and at home, 5.1 digital sound gave rise to a "digital surround style" (DSS) that affects both aural *and* visual aesthetics. Kerins notes the continued breakdown of classical Hollywood continuity rules that primarily dictate the visual aesthetics around widespread conventions of realism. Classical continuity ensured that time and space were clearly represented to the viewer through established visual cues such as the 180-degree rule, eye line matches, and establishing shots. Kerins argues that the increasing

prevalence of shaky cam, quick cuts, and seemingly incoherent editing strategies is a result of the affordances of digital surround sound to provide the temporal and spatial cues that were formerly the domain of the visual.[65] Because sounds can now accurately be placed in the surround speakers, digital surround sound is better equipped to indicate the spatial relationships between characters and objects between shots. Playing on Chion's superfield, Kerins calls the new digital soundscape the "ultrafield."[66] Where the surrounds in the superfield maintained a continuous omnidirectional soundscape, in the digital ultrafield, the *entire* soundscape shifts with *each visual cut*. Kerins points towards the opening invasion sequence of Steven Spielberg's *Saving Private Ryan* as a seminal example of digital surround style and the ultrafield in action.[67] As the soundtrack provides more precise spatial information, Spielberg can break continuity rules, and assume that the audience remains oriented by the soundtrack. In tandem with the surround speakers' placement around the auditorium, the increased dynamic and frequency range of digital surround style allows filmmakers to include more aural information on soundtracks with fuller sonic details helping to indicate place.

With the standardization of 5.1 digital sound for home and theatrical exhibition, digital surround style has remained the prevalent audiovisual aesthetic for Hollywood movies for the past two decades.[68] From the mono era to widescreen magnetic sound to Dolby Stereo to the

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[65] Kerins picks up David Bordwell's idea of "intensified continuity" in postclassical Hollywood film, but considers the role of sound, which Bordwell ignores in his analysis. Kerins concurs with Bordwell that contemporary Hollywood editing represents a continuation, not a break with classical continuity rules. See Kerins, *Beyond Dolby (Stereo)*, 118-23 and David Bordwell, *The Way Hollywood Tells It*, (Berkeley: U of California P, 2006).
[66] Ibid, 103-7.
[67] Ibid, 103-4.
[68] The past few years have seen the introduction of a number of digital formats that slightly modify the now familiar 5.1 speaker arrangement. The most common setup (used in some digital projection formats and Blu-ray) is 7.1, which adds two rear speakers in addition to the two

digital formats introduced in the early 1990s, audiovisual aesthetics slowly mutated according to the contingent material forces and industrial formations of the era. According to the logic of remediation, each new sound format is inevitably designed in relation to the technologies that preceded it.[69] Every cinematic audio system after monophonic optical sound was created to follow some basic overarching conventions of cinematic realism, but the audiovisual aesthetic associated with each format was not technologically determined. The superfield of Dolby Stereo and the ultrafield of digital surround were not forgone conclusions based on their respective technologies. These aesthetics developed over years of filmmakers experimenting with the affordances of the formats. Aesthetic strategies were sometimes shepherded by the creators of technology—as with Dolby consultants in the early days of Dolby Stereo—but no single format has been designed with such a narrowly prescribed set of aesthetic possibilities according to a singular mode of cinematic realism before Dolby Atmos.

The Ideology of Immersive Sound: Making and Making With Dolby Atmos

After a decade of digital sound accompanying analog film in theatrical exhibition, US distributors began transitioning to all-digital projection of sound *and* image with the 1999 release of *Star Wars: Episode I – The Phantom Menace*. Echoing his early advocacy of Dolby Stereo, director George Lucas heralded digital cinema as a revolutionary development in film history that would allow creators more freedom than ever before.[70] But because films were already being produced and edited digitally, the switch to digital projection for the image track did not automatically alter audiovisual aesthetics. Digital projection of *The Phantom Menace* arguably

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surrounds used in 5.1. The two additional channels in 7.1 do not significantly alter the digital surround style approach to soundtrack construction.

[69] Bolter, J. David, and Richard A Grusin. *Remediation: Understanding New Media*, (Cambridge, MA: MIT Press, 2000), 55.

[70] Lucas, George. "Movies Are an Illusion." *Premiere* 60 (February 1999), 60.

succeeded only as a remediation of film, not as a unique cinematic experience. As John Belton has observed, digital projection is not ontologically superior, or even noticeably different from 35mm analog film.[71] I don't wish to totally dismiss arguments made by those advocating for the unique auratic quality of film-as-film—but I do want to bracket questions of ontology and phenomenological reactions to digital cinema for the time being.[72] I instead want to focus on the significance of the move to digital *distribution*, which was not immediately apparent in the early techno-utopian praise for digital projection. Freed from the material constraints of 35mm film, which by 1999 was something of a palimpsest bearing traces of the decades technological history, studios could begin experimenting with a variety of new cinematic formats based on the affordances of relatively capacious digital distribution, from streaming to the Digital Cinema Package.[73] American studios' aggressive push for digital projection (and subsequent disavowal of film) set the stage for a variety of new cinematic formats, including the return of 3D imagery and various large screen approximations of IMAX.[74] Coinciding with the release of *The Phantom Menace*, Dolby introduced *Dolby Digital EX*, its first audio format designed for digital distribution. The 6.1 format adds a single rear center speaker in addition to the left and right surrounds in the name of "greater cinema realism."[75] With the release of *Toy Story* 3 in 2010, Dolby introduced Dolby Surround 7.1 (and its home video counterpart, Dolby TrueHD Lossless Audio for Blu-ray), adding yet another rear channel.[76] The rather involved 7.1 format ideally

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[71] Belton, John. "Digital Cinema: A False Revolution." *October* 100 (April 1, 2002), 104-5.
[72] For an overview of the negative reaction to digital projection among certain filmmakers and cinephiles see *savefilm.org*.
[73] Smith, "Atmos, All Around."
[74] Alimurung, Gendy. "Movie Studios Are Forcing Hollywood to Abandon 35mm Film. But the Consequences of Going Digital Are Vast, and Troubling." *LA Weekly*, April 12, 2012.
[75] "Dolby Digital Surround EX." *Dolby.com*.
[76] "Dolby Surround 7.1." *Dolby.com*.

places eight speakers (starting clockwise at the screen: front center, front right, surround right, rear right, rear left, surround left, front left, and subwoofer) around the spectator according to Dolby's spatial specifications.

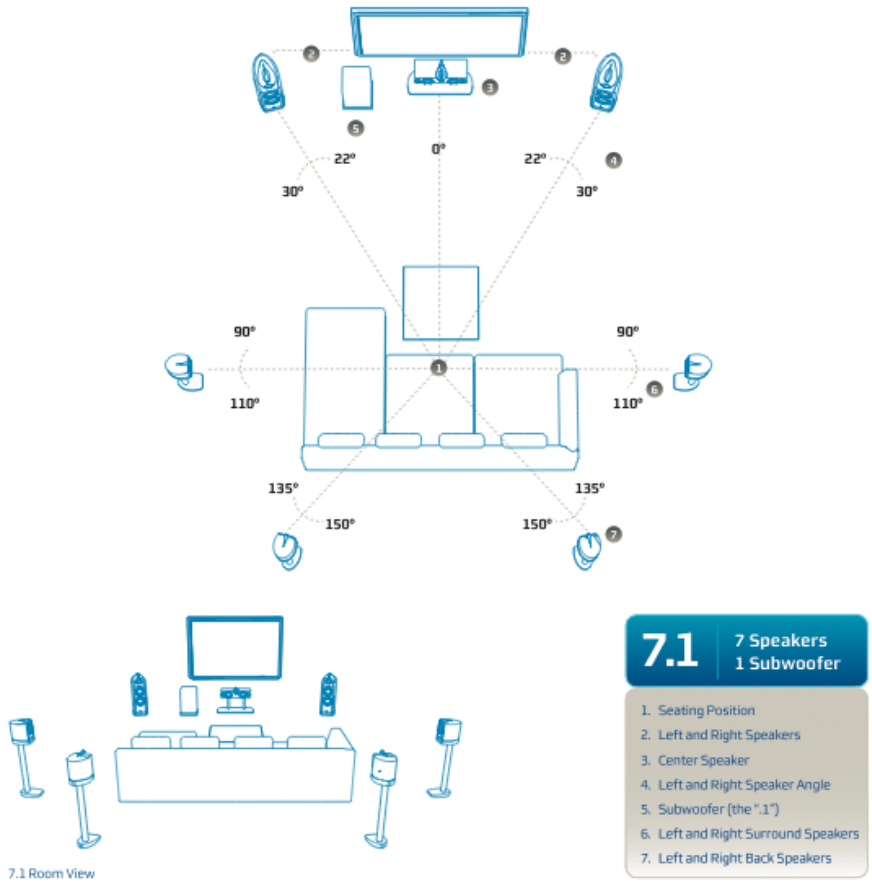

Figure 3. Dolby provides detailed instructions for the proper 7.1 speaker arrangement for home theater systems. "Home Theater Speaker Guide," Dolby Laboratories, 2010.

A decade after the introduction of digital projection and the Digital Cinema Package, Dolby had only incrementally altered the 5.1 audio standards created with Dolby Digital. The iterative addition of channels with EX and 7.1 was a relatively cheap and easy upgrade for cinemas to make, but a harder sell for home theater consumers. It is difficult for even home

theater enthusiasts to justify the expense and clutter of a 7.1 home theater system, considering the diminishing returns of additional surround speakers, the few titles (nearly all from the past 15 years) mixed for 7.1, and the paucity of formats (i.e. Blu-ray and a few specialty streaming services) that are capable of playing 7.1 mixes. Hampered by the entrenchment of 5.1 as a home format, simply adding additional channels was not a viable option in Dolby's development of new audio technologies to market to theaters and individual consumers. So when AMC Theaters approached Dolby to create a new sound format to compliment lucrative 3D films, simply adding more channels was not an option.[77] Taking advantage of the possibility of including increasingly complex metadata as a part of the Digital Cinema Package, Dolby designed Atmos to be the first post-channel cinematic sound format. Atmos is designed to be the final statement in what a Dolby executive calls the "channel wars," the iterative addition of one or more speaker channels to create a new format as with Dolby EX and 7.1. Atmos sidesteps the diminishing returns (and limited popularity) of these piecemeal improvements over 5.1 by dispensing with the concept of channels altogether.

This animating idea of moving beyond a channel-based format proves to be the most important distinction between Atmos and Dolby's previous sound formats. The particular speaker arrangement of the Atmos-equipped cinema or home theater (which I will get to shortly) is less significant to the future of audiovisual aesthetics than the format's innovative use of digital metadata afforded by the Digital Cinema Package. Traditionally, a sound designer would create a mix by placing sounds in the discrete channels provided by a given cinematic audio

[77] Wright, Benjamin. "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production." In *Living Stereo: Histories and Cultures of Multichannel Sound*, edited by Paul Théberge, Kyle Devine, and Tom Everrett, (New York: Bloomsbury Academic, 2015), 230.

format.[78] Every sound would have to be individually assigned to an individual speaker by the sound designer in order to create the soundtrack. To create a sense of movement through the soundscape, the sound designer must meticulously mix a sound into the appropriate speakers as it pans through the auditorium. For audio formats before Atmos, the ultimate channel arrangement is obviously predetermined, but the sound designer is free to use any technology or aesthetic conventions during production. As the development of digital surround style years after the introduction of digital sound illustrates, the technological form does not determine audio aesthetics. Although the vast majority of sound designers follow the spatially based realism conventions of digital surround sound, channel-based digital formats do not foreclose the possibility of further formal experimentation. The idiosyncratic 5.1 soundtrack to Jean-Luc Godard's recent *Goodbye to Language 3D*—which breaks usual spatial conventions by bringing attention to the discontinuous soundscape created by the five channels through erratic panning—demonstrates the still unexplored aesthetic freedom of channel-based digital sound.

Conversely, Atmos calcifies contemporary sound aesthetics by collapsing production and exhibition under Dolby's technically complex but "intuitive" system. In addition to its particular theatrical sound system, Atmos requires the use of Dolby's proprietary technology *during production* in order to create the soundtrack according to the format's "object-based" encoded meta-data. To distinguish Atmos from earlier and competing channel-based audio formats, Dolby has centered the system's marketing of the new aesthetic possibilities allowed by what the

[78] I recognize the industrial controversy over the title "sound designer." I use in its broadest sense to describe all activities involved in creating sonic space in audiovisual media. On the controversial nature of the title, see Whittington, *Sound Design and Science Fiction*, 26.

company is calling "audio objects."[79] These objects are essentially individual tracks (i.e. a single sound element created or recorded during production that is combined with other sound elements to create the final soundtrack) that are not assigned to specific channels during mixing.[80] Instead, the sound designer programs the object's movement through the soundscape using a purportedly intuitive joystick-like device.[81] The motion of the joystick is saved as metadata that controls the sound object's location in the speaker array depending on the number of speakers used for exhibition—from the minimum of ten speakers (recommended for home theater setups) to the maximum of 64 in the theaters.[82] The maximum of 128 unique audio objects exist alongside ten traditional channels, which Dolby has cannily renamed "beds" to underscore the novelty of the new format. Allowing sound designers the option to continue to use channels-cum-beds was intended to make transitioning to producing movies for Atmos as easy as possible—but Dolby representatives are careful to emphasize that beds are used only for static sounds, and claim that many movies are made only using objects.[83] The shift from the effectively manual mixing of channel-based formats to the algorithmic panning of Atmos is significant for two reasons.

First, the movement of audio around the soundscape is determined by Dolby to be the constitutive aesthetic feature of cinematic soundtrack. From the earliest days of stereophonic

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[79] The rhetorical move to concretize sound by describing it in the language of physical objects has a precedent in Christian Metz's essay "Aural Objects." Metz argues that cultural understandings of objectivity depend on the visual and tactile permanence of an object. Referring to sounds as objects was a way for Metz to discuss the aural in the primarily visualist language of film studies at the time. Created as a compliment to 3D cinema, Dolby makes a similar rhetorical move in describing sound in terms of physical objectivity. See Metz, Christian. "Aural Objets." In *Film Sound: Theory and Practice*, 154-61.
[80] "Dolby Atmos Next-Generation Audio for Cinema," 7.
[81] Sergi, "Knocking at the Door of Cinematic Artifice: Dolby Atmos, Challenges and Opportunities," 115.
[82] "Dolby Atmos Multidimensional Sound." *Dolby.com*.
[83] Wilkinson, Scott. *Dolby Atmos Questions Answered*. Home Theater Geeks.

sound, panning has always been a part of the stylistic toolkit that sound designers use to construct soundtracks, but Atmos situates it as the central element in both production and exhibition. In addition to the technology itself encouraging a certain audiovisual aesthetic, Dolby is careful to cultivate how the technology is employed. The first Atmos demonstration films were created for Dolby by famed sound designer Gary Rydstrom, who ultimately was the sound designer for *Brave*, the first feature-length movie released in the Atmos format.[84] In addition to hiring notable sound designers to demonstrate the capabilities of the format, Dolby offered financial incentives to studios to adopt Atmos in order to make upgrading more appealing to exhibitors. Echoing the company's tactic of providing gratis Dolby Consultants to work on early Dolby Stereo films, the company paid for studio time and helped steer the audiovisual aesthetics of early Atmos films including *Brave* (2012), *Life of Pi* (2012), *A Good Day to Die Hard* (2013), *Oblivion* (2013), *Star Trek Into Darkness* (2013) and *Gravity* (2013).[85] While an emphasis on movement in the soundscape might at first seem to imply that Atmos opens up possibilities for the soundtrack to do more independently of the images, in reality the increased spatial specificity more tightly sutures the sound and image together. As Michel Chion has noted, offscreen sound is always understood in relationship to the diegetic space depicted on the image.[86] So even as Atmos allows for more audio information, the increasingly spatially complex soundscape makes sense only in tandem, not in counterpoint, with the image. Acousmatic sounds can no longer have an ambiguous relationship to the screen, they must be placed in a specific location according to the specifications of Atmos's algorithmic mixing. Atmos takes the tight spatial

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[84] Wright, "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production," 227-8.
[85] Ibid, 238.
[86] Chion, *Audio-Vision: Sound on Screen*, 39-40.

relationship of 5.1 digital surround style to its logical extreme by emphasizing space and movement as the defining trait of audiovisual realism.

In addition to the greater prominence of movement and spatial specificity, the second significant aspect of the algorithmic mixing and panning in Atmos is the quiet codification of sound aesthetics according to the algorithmic standards of Dolby's proprietary technology. To loosely appropriate one of Henry Jenkins' key terms, Atmos is essentially a "black box" technology, in that it seems to transparently and intuitively mediate sound but is created according to complex cultural and technical specifications.[87] The history of cinema sound and Dolby's own rhetoric help elucidate the cultural formations that inform Atmos, but the specific algorithmic encoding that makes Atmos work is proprietary, and thus remains opaque as the "black box" metaphor suggests. Writing about text online, Alan Lieu has observed that the standardization of computer meta-languages such as XML effectively create a one-size-fits-all form that creators must make their content fit within. He argues that one of the defining features of postindustrial content creation is the separation between form and content necessitated by computer programs designed to standardize content delivery.[88] The world of audio production previously saw the introduction of a standardized meta-language in MIDI (Musical Instrumental Digital Interface), a computer allowing communication between synthesizers and audio equipment that created a certain homogeneity in music production.[89] But unlike the meta-standards of XML or MIDI, the code of Atmos is not customizable, or even observable by the

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[87] Jenkins, Henry. *Convergence Culture: Where Old and New Media Collide*, (New York: NYU P, 2008), 14-5.
[88] Liu, Alan. "Transcendental Data: Toward a Cultural History and Aesthetics of the New Encoded Discourse." *Critical Inquiry* 31, no. 1 (2004): 49–84.
[89] Pinch, Trevor. "Technology and Institutions: Living in a Material World." *Theory and Society* 37, no. 5 (July 10, 2008): 461–83.

content creators incentivized by Dolby to use it. For sound designers, Atmos is not a malleable format—it does not lend itself to sonic experimentation. Furthermore, the standardized computer metadata doesn't allow for sound designers to create a distinct sonic signature, or even to mix according to a more affective "dramatic feel" rather than Dolby's preferred spatial realism.[90] Atmos standardizes audio aesthetics, making the algorithm as much the sonic stylist as the sound designer by following the postindustrial logic that separates form and content.[91]

Despite the significance of the introduction of metadata and new production practices for the post-channel format, the number and placement of speakers continues to be the focus of Dolby's marketing push to exhibitors and excitement for the audiophile community. Atmos allows for up to 64 individual speakers in the auditorium (up from the previous maximum of eight for Dolby 7.1), adding more discretely controlled speakers around the auditorium and, for the first time for Dolby, on the ceiling.[92] Competing sound formats created by IOSONO and Auro actually introduced ceiling speakers slightly before Atmos, but Dolby quickly moved to make Atmos the most widely available immersive format.[93] For auditoriums installed with the maximum of 64 speakers, Atmos specifies the placement of five speakers behind the screen, nine speakers each along the right and left walls, six speakers in the rear, eighteen speakers on the ceiling in two rows, and two subwoofers placed on opposite sides of the theater.[94] Dolby refers to such an speaker setup as a 44.2.18 arrangement, with the first integer referring to the 44

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[90] Wright, "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production," 240-1.
[91] To a certain extent the bipartite structure of this document—split between examining the format of Atmos and textual analysis of Atmos movies—is meant to reflect this separation of form and content.
[92] Smith, "Atmos, All Around."
[93] Wright, "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production," 231.
[94] "Dolby Atmos Next-Generation Audio for Cinema," 13.

horizontally placed speakers, the second referring to the two subwoofers (following the numbering system of 5.1 surround), and the final number referring to the overhead arrays.[95] Each of the speakers operates individually only when a sound object pans through it or is statically placed in a specific location in the soundtrack. Otherwise, the speakers continue to operate as arrays according to the ten beds the format supports. Using Dolby's numbering system, these beds would be described as following a 7.1.2 arrangement, meaning the Atmos beds are essentially identical to the channels of Dolby 7.1, with the addition of two overhead speaker arrays. For home theater systems, the 7.1.2 speaker arrangement is the minimum required set up to play Blu-rays encoded in Atmos. Otherwise, the Atmos soundtrack automatically downmixes to a 2.0, 5.1, or 7.1 presentation depending on the home theater's audio capabilities. While 7.1.2 is the minimum for Atmos home presentation, the format encourages consumers to iteratively "improve" their home theaters with up to a theoretical maximum of 34 speakers.[96] While the metadata on Atmos-encoded Blu-rays can support up to 34 individual speakers, actually installing and powering that many is, so far, infeasible. The majority of high-end Atmos receivers are only designed to power the basic ten-speaker setup—only a handful support the addition of even a few extra speakers.[97] The possibility of continually upgrading the audiovisual arrangement of the home theater is the ultimate marketing coup in appealing to what Barbara Klinger calls the "new media aristocrats," who willfully conflate aesthetic discernment with the consumption of high-end home theater equipment.[98] Each additional speaker and receiver

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[95] Wilkinson, *Dolby Atmos Questions Answered*.
[96] "Dolby Atmos Multidimensional Sound." *Dolby.com*.
[97] "Dolby Atmos Home Theater Products." *Dolby.com*.
[98] Klinger, Barbara. *Beyond the Multiplex: Cinema, New Technologies, and the Home*. (Berkeley: U of California P, 2006), 33-9.

upgrade promises a richer aesthetic experience for those with the disposable income to invest in Atmos for the home.

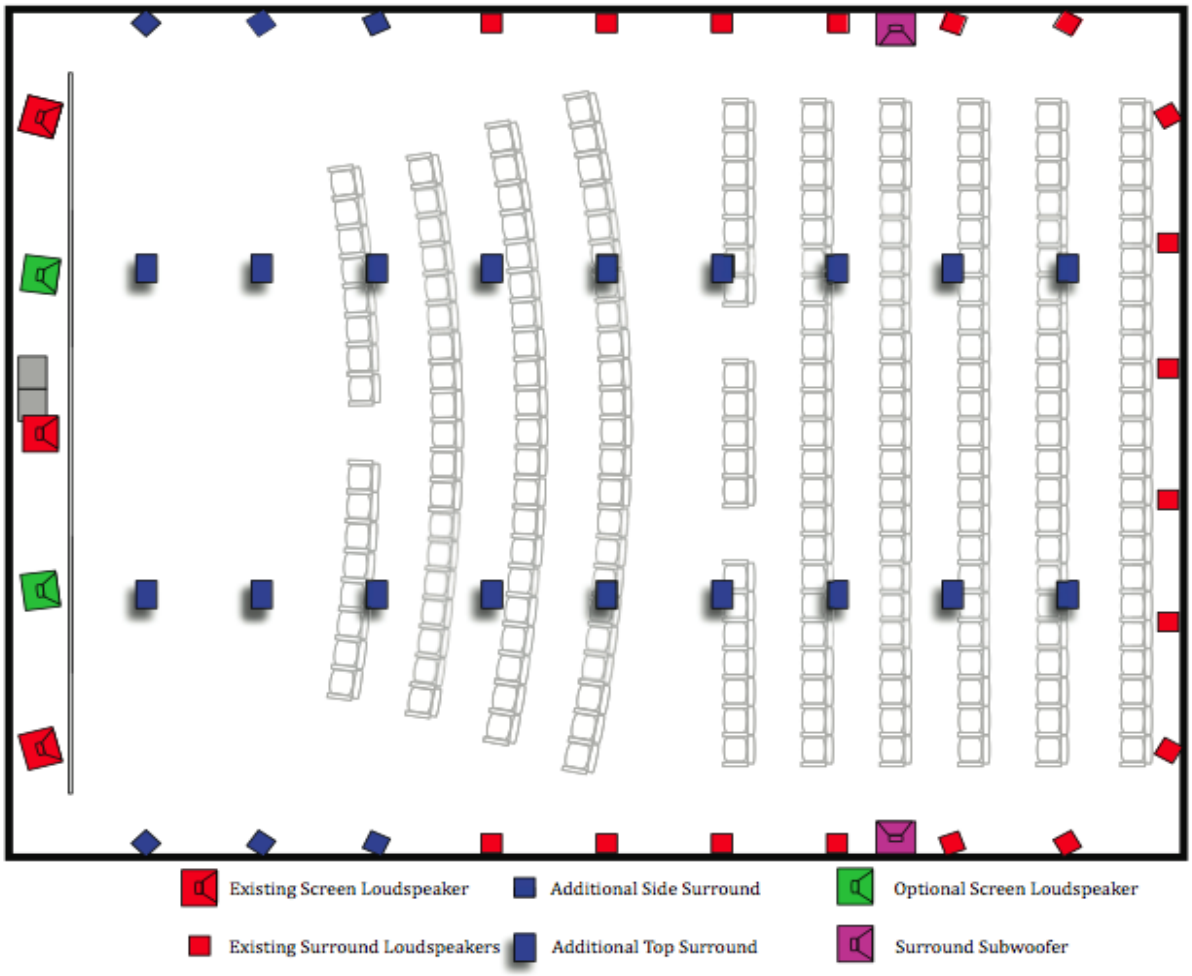

Figure 4. Dolby's specifications for a full 44.2.18 speaker arrangement. "Dolby Atmos Next-Generation Audio for Cinema." Dolby Laboratories, 2014.

In Dolby's early marketing material for Atmos, the ceiling speakers are a major selling point, and the primary justification for the "immersive sound" label.[99] Similarly, the third dimension in the "3D sound" designation used by competing formats such as Auro refers to the supposedly revolutionary addition of the overhead speakers to the previous horizontal sound

[99] Dolby Atmos Next-Generation Audio for Cinema," 4-5.

plane offered by 5.1 and 7.1 surround formats.[100] Yet despite the apparent importance of overhead speakers to immersive sound formats, it remains to be seen how and if vertical speaker arrays will affect audiovisual aesthetics over time. Furthermore, from a phenomenological perspective, it's debatable if an audience member would even notice the new overhead sound sources without Dolby prompting the development of a new audile technique. As Wilfried Van Baelen, the creator of the Auro-3D format acknowledges, human beings essentially hear on a horizontal plane, not a vertical one.[101] Overhead speakers arguably add extraneous sonic information that will not be immediately perceived as emanating from above. Recall that the very first shot of *Star Wars*, the veritable urtext of Dolby Stereo, depicts a spaceship passing over the audience as its accompanying sound panned from the surrounds to the front speakers. In this early surround film, the illusion of overhead movement is created only through sound on a horizontal plane. It's a fundamentally subjective proposition that the addition of overhead speakers to fill out this soundscape would create a more immersive, heightened sense of realism. Additionally, it's worth noting that overhead speakers are far from being a novelty in daily life. Sound and music are often pumped in from above in many banal institutional settings: conference rooms, restaurants, waiting rooms, and, of course, elevators. There is a certain irony in the rhetoric of disruption being thrown around to describe a speaker placement that is primarily associated with Muzak, background noise, and ultimately the effect of silence.[102]

After metadata and speaker arrangements, the final unique feature of Atmos is Dolby's requirement of an improved dynamic and frequency range in *all* of the speakers installed for

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[100] "Auro 11.1 - Barco's Immersive Cinema Sound Format," 2015.
[101] Wilkinson, Scott. *Auro-3D Immersive Audio*.
[102] Attali, Jacques. *Noise: The Political Economy of Music*. Translated by Brian Massumi. (Minneapolis: U of Minnesota P, 1985), 111-2.

theatrical exhibition.[103] Because the vast majority of sound emanates from behind the screen, the front speakers in most theaters have a significantly greater dynamic and frequency range compared to the surrounds. Dolby requires that all speakers in an Atmos theater have the same expansive dynamic and frequency ranges so that every speaker is technically capable of reproducing identical volumes and timbres. This requirement is partially in response to the poor standardization of cinematic sound across theaters.[104] As with most pushes for expanded frequency and dynamic ranges across media, Dolby equates more audio information with higher fidelity, and thus a greater mimetic realism. But cinematic sound, as I've already mentioned, is a studio art: a large portion of sounds are artificially created and placed in the soundscape through traditional foley practices or computer synthesis. The very structure of Atmos, with its reliance on algorithms to specifically place audio objects in the soundscape, necessarily forecloses the possibility of direct sound—there is no way for Atmos to capture the movement of actual sounds and sound objects on set. The original event that new speakers are ostensibly reproducing with high fidelity is the soundtrack as crafted by the sound designer. The increased dynamic range may accurately mediate the intentions of the filmmakers—which we've seen are generally tied up in historically and technologically contingent approaches to realism—but there is nothing definitively realistic in the speaker's technical ability to reproduce more audio information.

This observation on the relationship between realism and Atmos applies to the structure of the format as a whole. The algorithmic movement of audio objects, the specific spatial

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[103] Dolby Atmos Next-Generation Audio for Cinema," 5.

[104] For information on the lax sonic standards theatrically and the industry's recent push for more standardization in all theaters regardless of format, see Vessa, Brian, and Kurt Graffy. *Frequency Response Analysis of Theaters and Dubbing Stages*. (White Plains, NY: Society of Motion Picture and Television Engineers (SMPTE), 2014).

arrangement of the speakers, and the uniform dynamic and frequency response are, taken as a whole, only iterative improvements over previous audio formats according to the contingent rules of spatial realism that date back to the early years of sound cinema. This format, developed with input from an audiophile community of Hollywood sound personnel with an eye towards improving Dolby's cinematic revenue stream, reveals more about the sound cultures of American cinema than it does about human phenomenology and perception. The seeming disjunction of this supposedly realistic sound format being primarily associated with the fantastical genres of science fiction and action begins to make sense when you understand that Atmos was really created to mediate *spectacle* according to conventions of realism, not to mediate reality itself.[105] Thus far I've mostly focused on how Dolby has positioned Atmos in relationship to filmmakers, exhibitors, and a relatively hermetic group of audiophiles interested in acquiring the latest cinematic technology. Turning attention to the actual motion pictures made to market Atmos to a general theater-going audience—whose appetite for the format will ultimately determine its success—offers a perhaps the clearest illustration of the underlying ideological assumptions of Atmos, and what Dolby and filmmakers themselves see as its primary appeal to a public that is mostly indifferent to the technical specifics of film sound.

[105] This conception of Atmos (and by extension earlier sound formats) shares certain similarities with Jean Baudrillard's concept of the "hyperreality." I woudn't go so far as to say that immersive sound necessarily severs the spectator's relationship with reality, only that Atmos confuses the boundaries between the fantastic and the real. Baudrillard doesn't get into the specifics of the sound presentation, but it's significant that he singles out early Dolby Stereo film *Apocalypse Now* as one of his main examples of the hyperreal in the cinema. See Baudrillard, Jean. *Simulacra and simulation*. Translated by Sheila Faria Glaser. (Ann Arbor: U of Michigan P, 1994).

CHAPTER 2: THE SOUND OF SCIENCE (FICTION): LISTENING TO ATMOS

After nearly three years and over 150 films released with Atmos soundtracks, it's possible to begin making some tentative observations about the audiovisual aesthetics of immersive sound. Despite my contention that Atmos effectively forecloses the development of formal strategies that run counter to the conventions of digital surround sound, formal analysis is necessary to fully understand the limits and possibilities of the format. Rick Altman's idea of crisis historiography, which posits that representational technologies are only fully understood through cultural practice and "are not fully and self-evidently defined by their components and configurations," is surely relevant to the particular technological instantiation that is Atmos.[106] Although the format was carefully created to adhere to specific conventions, Dolby cannot fully determine the aesthetics of immersive sound—only filmmakers can. Observing how creators accept, resist, and circumvent the strictures of Atmos is of primary concern in the following readings.

As important as formal analysis is, there are numerous obstacles that make studying an Atmos soundtrack a difficult undertaking compared to earlier formats. First and foremost, there is the issue of scarcity. In this age of media ubiquity, the difficulty of simply hearing the Atmos soundtrack for a given movie can feel like a relic of the bad old days of a film's theatrical run being the only opportunity to view it. Only a small percentage of Atmos soundtracks have been released on home video, and very few theaters support the format. There is only one Atmos screen in the Washington, DC metro area, and there is no guarantee that the theater chain will

[106] Altman, Rick. *Silent Film Sound*. (New York: Columbia UP, 2004), 16.

always play an Atmos release in the equipped auditorium.[107] Even when Atmos titles are released on Blu-ray, the very nature of the post-channel format makes studying the soundtrack very difficult. With a discrete number of analog channels, it's relatively easy to study the how the sound designer assembled various components to create the soundtrack. Blu-ray or DVD players with analog outs can be run through a relatively inexpensive mixer in order to isolate and closely observe the soundtrack channel by channel. Because of the necessary metadata involved with Atmos soundtracks, digital HDMI cables must be connected directly to the receiver instead of analog outputs. Manually disconnecting and reconnecting individual speakers one-by-one is the only way to isolate and study individual soundtrack elements.

Finally, whether in the theater or the home, it's difficult to point to the definitive Atmos soundtrack. Unlike the image, which thanks to the efforts of home video companies like the Criterion Collection is always presented in a set aspect ratio, the soundtrack of a film is surprisingly mutable.[108] Canonical films are routinely restored with noticeably different sound mixes according to the prevalent technology of the time. This ontological uncertainty is heightened in the current production climate, where a 5.1 or 7.1 mix is created in tandem or before the Atmos mix.[109] Having many iterations of the same soundtrack is accepted as the norm

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[107] The coincidental release of the *Interstellar* (in 5.1) and *Big Hero 6* (in Atmos) in summer 2014 meant that the Atmos mix of the latter film was never screened in Washington—*Interstellar* was shown in the Atmos auditorium, which not coincidentally features the largest screen in the multiplex. Similarly, the close release of *Kingsmen: The Secret Service* and *McFarland, USA* (both in Atmos) led to the latter film never being screened.

[108] Restoration practices for soundtracks are wildly inconsistent. For a case study on the reconstituted soundtrack of *Vertigo* that quietly became the definitive version despite being created nearly 40 years after the film's theatrical release, see Kilcoyne, Sean. "You Shouldn't Have Been That Sentimental." *Journal of Information Ethics* 19, no. 1 (April 1, 2010): 57–73.

[109] There is even debate over what qualifies as the first Atmos release. *Brave* was upmixed to Atmos after the Dolby 7.1 was created. The 2012 Tom Cruise vehicle *Oblivion* is widely reported to be the first film mixed natively for Atmos. Giardina, Carolyn. "'Oblivion,' 'The

for current Hollywood releases, making the identification of the definitive version essentially impossible. Furthermore, because Atmos uses metadata to place audio objects in approximate locations according to the specific number of speakers in the exhibition space, it's debatable if the Atmos soundtrack itself is a fixed object. An argument could be made that every Atmos screening is formally unique depending on the speaker arrangement. For the purpose of my formal analyses, I'm going to bracket these particular questions and acknowledge that my analysis any Atmos soundtrack is inherently based on my subjective, contingent listening experience. Also, I stipulate that regardless of whether an Atmos mix was made before, after, or at the same time as a 7.1 or 5.1 soundtrack, the resulting audiovisual work can and should be understood ontologically as an Atmos movie when viewed in the format. In this I privilege the spectator's phenomenological experience of the Atmos soundtrack and its interplay with the image track when analyzing movies. To get a complete sense of Atmos aesthetics, I focus on moments of tension *and* congruity—I am interested in how filmmakers alternately chafe against or embrace the design of the format, often within the same movie. But before turning to feature-length films, it's important to first examine the short promotional trailers created by Dolby in conjunction with professional filmmakers. These shorts illustrate Dolby's ideal application of Atmos, and serve as examples or templates for sound designers to refer to when creating new soundtracks.

Immersive Marketing: The Dolby Atmos Trailers

The short promotional film "Leaf," made in collaboration with Dolby, Pixar, and sound designer Gary Rydstrom was the first major public introduction to Atmos when it played before

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Croods,' 'Monsters University,' 'Iron Man 3' Will Get Atmos Treatment - Hollywood Reporter." *The Hollywood Reporter*, February 14, 2013.

Pixar's *Brave* in the handful of equipped theaters in 2012.[110] The anodyne computer-animated short begins with onscreen text asserting simply: "At the heart of great cinema is great sound."[111] It goes on to depict a dried "helicopter" seedling—Rydstrom's playful allusion to Walter Murch's celebrated helicopter sound design in *Apocalypse* Now—falling from a Maple in a lush forest to eventually land in a pool of water that reveals the Dolby Atmos logo.[112] As the seedling makes it erratic descent, the camera swoops in and out of the foliage, occasionally losing sight of it in order to allow the attendant audio object to move throughout the theater's soundscape, unmoored from the screen. For those attuned to closely listening to cinema sound, the short is impressive in its meticulous spatial placement of the sound object as it seems to pass through and around the auditorium. But as an audiovisual artifact it is far from captivating. A short showing a seedling falling from a tree can only be so exciting, no matter how painstakingly its visuals and soundtrack are rendered. As Atmos expanded to other theaters in the following years, "Leaf" would prove to be the template for Dolby's promotional trailers, which singularly focused on scenes and sounds of nature to highlight the format's propensity for the accurate mediation of reality. Comparing the Atmos trailers to promotional shorts for previous Dolby formats underscores the realist thrust of the marketing for immersive sound.

In her essay "When the Ear Dreams: Dolby Digital and the Imagination of Sound," Vivan Sobchack offers close readings of the nine trailers made from the1990s to 2003 in order to

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[110] Wright, "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production," 227.
[111] *Dolby Atmos: "Leaf" Cinema Trailer*. Youtube, 2014.
[112] Wright, "Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production," 228.

promote the Dolby Digital format both on home media and theatrically.[113] The early Dolby Digital trailers utilize stylized computer animation to depict a variety of unreal cinematic spaces that eventually reveal the familiar Dolby logo.[114] The first of these trailers, "Train," begins with a black screen accompanied by the familiar sounds of a steam engine slowly picking up speed. The train eventually appears as a ghastly, translucent computer generated image that implies a digital reimagining of the seminal Lumiére Brothers film, "The Arrival of a Train at La Ciotat Station." Other shorts depict similarly phantasmagorical images of a city, temple, and canyon accompanied by the precise digital soundtrack. Sobchack argues that these trailers dispense with the usual literal synchresis between sound and image to explore a more oneiric, metaphorical relationship between sound and image in which sound is dominant.[115] Her readings of the early Dolby Digital trailers suggest that in the early days of digital sound, Dolby and theatrical audiences were enamored with the imaginative possibilities of the new format and digital rendering in general. Digital sound's pairing with computer animation implies the creative freedom of digital tools to make new audiovisual creations outside the limitations of implicitly indexical analog media.

However, Sobchack's suggestion of a new, allegorical relationship between sound and image breaks down in her consideration of the final Dolby Digital trailer, "Stomp." Unlike the

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[113] These dates correspond with the years of Dolby Digital's market dominance, expanding from 800 screens in 1995 to 35,000 in 2003 thanks in large part to the theater construction boom of the 1990s. See Natale, Richard. "Company Town: Three Companies Wage Battle for the Hearts--and Ears--of U.S. Moviegoers." *Los Angeles Times*, September 5, 1995. and Kerins, *Beyond Dolby (Stereo)*, 44.

[114] Dolby's official YouTube channel does not currently host any of the early Dolby Digital trailers, but they are readily available on other users' channels. User surroundFM uploaded the majority of these trailers to YouTube in 2010. See https://www.youtube.com/channel/UChSCz_2GD4WRMvjW2ceQWpA

[115] Sobchack, Vivian. "When the Ear Dreams: Dolby Digital and the Imagination of Sound." *Film Quarterly* 58, no. 4 (June 1, 2005): 15.

earlier animated trailers, this filmed short features the titular dance group in a conventionally synchronized audiovisual display of their trademark idiosyncratic percussive performance. Made in 2003, the "Stomp" trailer is notable for having been produced as the conventions of digital surround style were increasingly codified as the wake of 5.1 movies like *Saving Private Ryan*, *The Matrix*, and *Gladiator*.[116] The transition from the oneiric "Train" to the literal "Stomp" reveals Dolby's dawning awareness of the strict spatial regime of Dolby Digital and a resulting commitment to the conventions of cinematic realism. These trailers reveal Dolby's waning interest in the imaginative and spectacular possibilities of new digital technologies. The imaginative and unreal aspects of the technology eventually succumb to the strictures of realism of the later "Stomp" advertisement. This ideological shift is instructive in understanding the later Atmos trailers like "Leaf" that stress the mimetic potential of the format.

After "Leaf," Dolby produced a handful of other trailers that were more widely seen as Atmos expanded to more theaters. "Amaze" is a thematically similar video, but ditches Pixar's realistic renderings of a forest in favor of hyper-detailed video recordings of various nature scenes reminiscent of the HD cinematography of the BBC documentary series *Planet Earth*. This not-so-subtle visual allusion to the verisimilitude of the popular documentary series is accompanied by overlaid text proclaiming the arrival of "The future of cinema sound" that is "natural and lifelike."[117] Each shot is accompanied by a mimetic soundscape highlighting Atmos's capabilities: rain falling from the overhead speakers, the rustling of branches in the surrounds, and the rumbling of thunder in the LFE channels. Around the same time as "Amaze," Dolby released the comparatively lighthearted "Conductor" trailer to accompany Atmos movies

[116] Kerins, *Beyond Dolby (Stereo)*, 84-111.
[117] *Dolby Atmos: "Amaze" Cinema Trailer*, 2014.

marketed to children. With Pixar-like computer animation by Moonbot Studios, "Conductor" features a young girl in a conductor's tuxedo facing the audience.[118] The girl appears to point her conductor's baton at various spots in the auditorium, with various wildlife sound effects emanating from the point she gestured towards, as if she was conducting an orchestra of animals. The recordings of apparently real animals contrasted with the clearly artificial animation of the little girl imply that Dolby Atmos is mimetically superior to accompanying image track. The animated image track cannot compete with the immersive reality of Atmos. "Leaf," "Amaze," and "Conductor" all rely in some part on the sounds of nature in an effort to bring attention to the fundamental natural realism that Atmos offers.

Recently—perhaps as an attempt to course correct the sluggish adoption of Atmos—Dolby released another trailer, "Unfold," that is more reminiscent of the surreal imagery and soundscapes of the early Dolby Digital trailers. Clearly inspired by the audiovisual aesthetics of the *Transformers* movie series, "Unfold" depicts an amorphous computer-animated crystalline blob that pulsates on screen before breaking into numerous shards that disappear from the screen to travel around the offscreen soundscape. The audio objects representing the individual shards ultimately reconstitute into a geometric shape that is eventually revealed to be the Dolby logo. The trailer ends with a new tagline, "Feel Every Dimension" showing Dolby's apparent hedging on their commitment to pure realism by gesturing towards subjectivity and affective response. The "Unfold" trailer debuted in late 2014, so it remains to be seen if Dolby will attempt to alter how it markets Atmos to filmmakers and audiences alike. Such a shift realist to affective

[118] Moonbot Studios also created the short film "Silent" for Dolby, which celebrates the role of science and technology in the process of moviemaking.

aesthetics will be difficult considering the very structure of Atmos, which was ultimately created according to the strict spatial rules and mimetic ideology of digital surround sound.

Remediating Space: James Gunn, David Acord, and the Audiovisual Aesthetics of *Guardians of the Galaxy*

As a high-grossing, critically praised spectacle, *Guardians of the Galaxy* is a perfect example of the very type of movie that will ultimately determine the success of Atmos as a format. For immersive sound to displace surround sound as the dominant theatrical format, exhibitors must be convinced that audiences demand the more expensive Atmos experience. As an advertisement for the possibilities of Atmos, *Guardians of the Galaxy* is a particularly fraught text with many overt stylistic antecedents that guide its audiovisual aesthetic. The movie is a palimpsest of earlier sound styles and technology that both textually and implicitly forces the Atmos soundtrack to remediate the form and content of preceding formats.

Based on an obscure Marvel comic book, *Guardians of the Galaxy* is light comic space opera in which a ragtag group of heroes face off against an evil galactic empire. The movie is obviously indebted the filmography of George Lucas. The plot, tone, and audiovisual aesthetics owe much to the Dolby Stereo-era *Star Wars* films, which like *Guardians of the Galaxy* and Marvel Studios is currently under the Disney corporate umbrella. This *Star Wars* connection is made clearer by the involvement of sound designer David Acord, who prior to working on *Guardians of the Galaxy* worked on the soundtracks of a variety of *Star Wars* proejcts. Furthermore, *Guardians of the Galaxy* features a soundtrack of familiar popular songs that emanates from within the diegesis, as in Lucas's 1973 film *American Graffiti*. Director James Gunn's textual foregrounding of sound technology in the protagonist's cassette player that ostensibly provides the film's soundtrack brings attention to the movie's use of the Atmos format

in mediating the diegetic pop songs. In addition to the connections to George Lucas's filmography, *Guardians of the Galaxy* is a part of the mega-franchise of films produced by Marvel Studios that take place in the so-called "Marvel Cinematic Universe." Marvel producer Kevin Feige is notoriously involved in maintaining a consistent style across all Marvel movies, meaning that the filmmakers can only make limited modifications to the prescribed audiovisual style developed in pre-Atmos Dolby Digital films such as *Iron Man* (2008), *The Incredible Hulk* (2008), *Iron Man 2* (2010), *Captain America: The First Avenger* (2011), and *Marvel's The Avengers* (2012).[119] Instead of examining the Atmos soundtrack of *Guardians of the Galaxy* in terms of how it remediates the style and content of stereophonic pop music, Dolby Stereo space opera, and Dolby Digital comic book movies, I want to contextualize the movie within the careers of the filmmakers who are most directly responsible for its audiovisual aesthetics: director James Gunn and sound designer David Acord.

Before *Guardians of the Galaxy*, James Gunn directed only two movies: the scifi-horror-comedy *Slither* (2006) and the black comedy superhero deconstruction *Super* (2010). Gunn's playful genre mixing of science fiction, superheroes, and comedy runs through his filmography, along with the prominent diegetic use of popular songs and the attendant depiction of media technology. In terms of style and content, *Guardians of the Galaxy* is similar to Gunn's 2006 debut. *Slither's* rural setting and zombie-like contagion plot are more closely aligned with low-budget genre experiments like *Evil Dead* (1981) than with Lucas's ornate space operas, but the movie has a few moments that build on audiovisual tropes introduced in *Star Wars*. Like

[119] Kevin Feigie has positioned himself as the auteur of the Marvel Cinematic Universe, dismissing directors like Edgar Wright who deviate too much from Marvel house style. See Child, Ben. "Ant-Man: Kevin Feige Rejects Marvel Criticism over Edgar Wright Exit." *The Guardian*, July 16, 2014.

countless other movies that involve extraterrestrial life, *Slither* borrows the opening shot of *Star Wars*, having the sound and image of an object in space pan from the rear to the front speakers as it seems to pass over the audience towards a planet below—a ready example of a Dolby Stereo technique being carried over to Dolby Digital. *Slither* also features a few brief glimpses in flashback of the home world of the invading aliens. These brief shots fill the surround speakers with fantastic sounds of alien life forms reminiscent of Ben Burtt's iconic otherworldly sound effects that have been prominently used in countless iterations of *Star Wars* media. *Super* has a less immediately arresting soundtrack, primarily making due with familiar action movie tropes, such as the exaggerated foley effects during the would-be hero protagonists physical altercations with prisoners. Throughout, both movies adhere to Mark Kerin's conception of digital surround style, with a shifting sonic field that matches point of audition and point of view with each visual cut. In terms of his approach to sound effects and spatial realism, Gunns work—up to and including *Guardians of the Galaxy*—is decidedly typical of contemporary filmmaking practices.

As a filmmaker who got his start writing screenplays (his credits include *Scooby-Doo* (2002) and *Dawn of the Dead* (2004)), Gunn seems to be interested more in playing with tone and genre in his scripts than in experimenting formally with sound and image. Watching the movie theatrically at two cinemas equipped for 5.1 sound and Atmos respectively, there were few perceptible differences in the soundtracks mixed for the different formats.[120] The Atmos mix was slightly clearer (I heard all of the dialogue in Atmos that was occasionally masked in the 5.1 mix), but on the whole it was essentially a louder, more detailed version of the Dolby Digital soundtrack that followed familiar realism conventions. During the two aerial set pieces—in the middle of the film as the heroes navigate space pods outside the orbit of a black-market hub

[120] The Atmos theater I attended in the DC suburbs featured a 27.2.18 speaker arrangement.

called Knowhere, and in the climactic battle above the peaceful planet Nova Prime—the audio objects pan through the soundscape nearly identically in the Atmos and 5.1 mixes. Unlike other Atmos mixes for science fiction movies (which I will turn to shortly) *Guardians of the Galaxy* makes no extra effort to stage action scenes that will bring attention to the new ceiling speakers. Except for a few explosions, scattered gunfire, and the destructive field of energy that surrounds the heroes as they come together to keep the McGuffin "infinity stone" from the power-hungry villain Ronan, audio objects rarely move across the vertical field, which seems to indicate Gunn's indifference to the Atmos mix compared to the more widely heard surround mix.

Despite the conventional audiovisual aesthetics that characterize his work, Gunn demonstrates an interest in sonic mediation that runs through his three films. This depiction of audio technology becomes a key plot point in *Guardians of the Galaxy*, but is evident in lesser ways in his earlier films. With obvious inspiration from Quentin Tarantino, Gunn makes use of pop songs—usually light rock recordings from the 1970s and 80s—to create a tonal disjunction between sound and image. Gunn has the sound of the music ambiguously shift between emanating from a diegetic source and playing outside of the diegesis, further heightening a sense of unease about the recording's relationship to the scene. In *Slither*, this happens near the climax of the movie as a grotesque man-monster infected by an alien virus plays a recording of Air Supply's 1980 single "Every Woman in the World to Me" in a bizarre attempt to romantically entice a victim. Before the viewer is shown the diegetic sound source, the song is heard from the surrounds, apparently indicating that it is a nondiegetic part of the soundtrack. But as the victim approaches the sound's source within the diegesis, the relationship between sound and image becomes tighter, until it is clear the song is emanating from a CD player that begins skipping after being accidentally bumped by an onscreen character. Gunn employs this strategy in reverse

in *Super*, when Swedish rock band The Ark's song "Let Your Body Decide" shifts from emanating from a diegetic home stereo to filling the surround channels, occupying a semi-diegetic space. In a monophonic film, such spatial shifts between the front channels and the surrounds would be impossible—with all sound emanating from one speaker, the music would ultimately be understood as diegetic throughout. The 5.1 speaker arrangement of Dolby Digital allows for subtle shifts between diegetic and non-diegetic sound without drawing undue attention to the uncanny breaking of spatial conventions. Although they are often used to impart spatial information, the two surround channels can also serve as an indefinite non-diegetic space that is difficult to convey with the hyper-specific speaker placement of Atmos as Gunn's use of pop music in *Guardians of the Galaxy* demonstrates.

After being abducted from Earth by a ship of space bandits, *Guardians* protagonist Peter Quill is sentimentally attached to his prized Walkman and one cassette tape, *Awesome Mix Vol. 1*, which is the last connection he has to his home planet and his deceased mother. Quill is at times irrationally protective of his Walkman, inciting a number comic action sequences in trying to protect or retrieve it.[121] The opening scene of the movie echoes the "Every Woman in the World" sequence of *Slither*, as the enveloping, seemingly nondiegetic music of 10cc's "I'm Not in Love" is revealed to be emanating from the young Quill's headphones as he sits in a hospital waiting room. In the next scene, fully grown Quill makes his way through a hostile alien world while listening to Redbone's "Come and Get Your Love" on his headphones. Later in the film,

[121] The mixtape's central place in the plot coupled with the movie's massive success improbably led to the official commercial release of *Awesome Mix Vol. 1*—a collection entirely comprised of previously released and readily available songs—to reach #1 on the US Billboard Album chart in 2014. See Caulfield, Keith. "Billboard 200 Chart Moves: 'Guardians' Soundtrack Surpasses 1 Million in Sales," *Billboard.* January 16, 2015.

other 1970s pop songs are heard playing from the tape deck in Quill's spaceship, shifting between diegetic and nondiegetic space on the soundtrack. In the Atmos mix of *Guardians of the Galaxy,* such shifts are particularly jarring. When the music transitions from a specific diegetic placement in the speaker array to an ambiguous nondiegetic or metadiegetic placement, the song's audio object expands to play out of every speaker in the auditorium. For an audience member primed for the careful spatial placement and naturalistic mediation that Atmos promises in the "Amaze" trailer accompanying the picture, the expansion of the 2.0 stereo pop song to the dozens of speakers surrounding the auditorium is surreal.

As a rule, in the Atmos mix for *Guardians of the Galaxy*, sound objects with no specific diegetic location are mixed to all channels. There are two categories of such 'non-locational' sounds: diegetic, encompassing ambient sound, the background hum of a spaceship, or general room tone; and non-diegetic, which is primarily the pop music of *Awesome Mix Vol. 1* along with some instrumental scoring by Tyler Bates. While omnidirectional ambient sound is not immediately discernable, the experience of being totally enveloped by loud music from the surrounds and from above is disorienting, having no correspondence with familiar presentations of recorded music intended for two-channel stereo presentation. This unreal mediation of ordinary content highlights the delicate relationship between the technologically sophisticated Atmos system and the everyday experience of sonic reality that the format is supposed to transparently mediate. The presence of the incongruous outdated Walkman technology and half-remembered radio hits in the otherwise futuristic world of *Guardians of the Galaxy* thematizes this disjunction. Through Quill, the movie privileges the familiar, authentic outdated technology of the Walkman and cassette tape at the expense of the uncanny alien technology that is both represented by and associated with the Atmos system. The very presence of the Walkman and

cassettes within the diegesis, and their fundamental incompatibility with Atmos defamiliarizes conventional representational strategies and calls into question the relationship between Atmos and the real that it supposedly mediates.

Along with Gunn, sound designer David Acord must be recognized as the author of the soundtrack. In addition to his work on *Guardians of the Galaxy*, Acord has had an impressively varied career, working on sound with a variety of directors including Paul Thomas Anderson, J.J. Abrams, and Kevin Smith. Perusing his many credits, it becomes apparent that he was likely chosen to helm the sound design of an expensive tent-pole blockbuster because of his work on a number of *Star Wars* properties. Starting with *Star Wars: Episode II – Attack of the Clones* (2002), Acord has worked in some capacity in every major *Star Wars* release, including *Star Wars: Episode III – Revenge of the Sith* (2005), the animated spinoff movie *Star Wars: The Clone Wars* (2008), and two recent television series produced by Disney: *Star Wars: The Clone Wars* (2009-14) (which continues the story of the identically titled motion picture) and *Star Wars Rebels* (2014-15). Acord's *Star Wars* film work adheres to the digital surround style put in place by the first prequel and first Dolby EX release, *Star Wars: Episode I – The Phantom Menace* (1999). The soundtrack for that movie was overseen by original *Star Wars* sound designer Ben Burtt, who tweaked the Dolby Stereo superfield aesthetics of the original trilogy to be more in line with the ultrafield conventions of late 1990s digital sound. Working with Burtt on the latter two prequels allowed Acord to master the audiovisual style of Lucas's universe that he later brought to *Star Wars* spin-offs and *Guardians of the Galaxy*.

The feature films Acord worked on are so heavily indebted to Burtt's sound design that it's difficult to appreciate what unique touches Acord brings to the table. The cinematic soundtracks are over-determined by the spatial conventions of digital surround sound and the

audiovisual aesthetics of the original trilogy. The *Star Wars: Clone Wars* television show, on the other hand, reveals more about Acord's particular sonic approach as he transports the franchise across other media. The first five series of the show were aired on Cartoon Network until Disney purchased Lucasfilm and the rights to *Star Wars*. As Disney prepared to release a new series of *Star Wars* films, it chose to cancel the series in order to focus on marketing and branding the new cinematic content. After Disney pulled the show from Cartoon Network, episodes from the yet to be aired sixth season of the series were completed and released on Netflix in 2014.[122] To better understand the unique aesthetics of Atmos, it's useful to observe Acord's approach to sound design shifts between the cinema with *Guardians of the Galaxy* to various seasons of *Star Wars* intended for both television and online, where the viewer will likely be encountering the film in two-channel stereo. In terms of its presentation plot and characters, the series is completely inaccessible without substantial familiarity with the mythos of the *Star Wars* universe. The initial episodes of this apparently standalone series makes no effort to acquaint viewers with the many characters introduced and characterized in other media. Throughout its run the series feels more like an ancillary tie-in than a discrete work to be enjoyed on its own terms.

This sense of supplementation and a reliance on preexisting content to gin up interest carries through to the series' formal attributes. Tellingly, episodes are presented in a widescreen 2.35:1 aspect ratio of the *Star Wars* films, not the usual 1.77:1 (or 16:9) aspect ratio that is the standard for HD televisions. Visually, the show makes use of the stylized computer animation introduced in the *Star Wars: Clone Wars* feature film that represent the characters, vehicles, and places as they appeared in the *Star Wars: Episode II*. Aurally, the series makes use of Ben

[122] Dave Itzkoff, "Clone Wars Moves to Netflix," *The New York Times*. February 13, 2014.

Burtt's iconic sound effects that have come to signify *Star Wars* as much as the franchise's visual iconography. However, Acord does not employ the same digital sound ultrafield aesthetics used in the *Star Wars* prequels and the *Clone Wars* feature film. In deference to the likely 2.0 stereo presentation of most televisions and computers, Acord does not bother to make the *entire* soundscape change with every visual cut. Because most people will not hear the surround channels, they are not heavily relied upon to provide additional spatial cues. Acord reverts to Dolby Stereo superfield aesthetics, keeping the ambient sound from the surround channels consistent within a scene consistent regardless of cuts that would call for a different point of audition.[123] The "retro" audio approach of the *Star Wars: Clone Wars* series points to the vestigial survival on air and online. This means that sound designers like Acord are consciously choosing to employ 5.1 digital surround sound aesthetics in their feature film work. The use of the ultrafield is a specific stylistic choice, not a reflexive, compulsory action. Acord's work displays a degree of aesthetic variety, meaning that the digital surround style used for *Guardians of the Galaxy* reflects the sound designer's understanding of Atmos as just an extension of digital surround style, not a paradigm shift for cinematic sound. The fact that Atmos does not engender a shift in aesthetic strategy demonstrates the format's essential fealty to the spatial realism of digital surround sound.

[123] This indifference to the surround channels and the reversion to Dolby Stereo-era superfield aesthetics is prevalent in the 5.1 mixes for most television series and many movies that can count on a VOD release, but face fewer options for theatrical exhibition. Recent independent movies by young filmmakers demonstrate an indifference to the surround channels. Director Charlie McDowell's 2014 scifi drama *The One I Love*, for example, features noticeably inferior recordings of ambient noise in the surrounds that seem to loop in longer scenes.

Other Atmos Releases: Michael Bay and Beyond

Guardians of the Galaxy is a particularly interesting case study of Atmos aesthetics because of its critical and commercial success, its intertextual relationship to earlier science fiction films, and its foregrounding of issues of sonic mediation. But focusing on only one feature film gives an incomplete picture of the variety of ways that sound designers are approaching Atmos soundtracks. In order to get a fuller sense of Atmos aesthetics, I want to briefly examine the first four Atmos titles released for home video on Blu-ray. These movies, all released theatrically in 2014, represent Dolby's earliest forays into the home theater market, which will be crucial for the widespread adoption of Atmos as a cinematic standard. Two of the movies, *Transformers: Age of Extinction* and *Teenage Mutant Ninja Turtles*, are Michael Bay productions that fall into the scifi/action genre. The other two film fall outside of the realm of science fiction, but provide worthwhile counterpoints to the understanding of the relationship between Atmos and scifi aesthetics. The Sylvester Stallone vanity project *The Expendables 3* is a action movie. The final film, *Step Up: All In* is the fifth entry in the long-running dance series with roots in the Hollywood musical tradition.

With action-spectacle auteur Michael Bay serving as a producer for both films, *Transformers: Age of Extinction* (the fourth entry in the *Transformers* series, directed by Michael Bay himself) and *Teenage Mutant Ninja Turtles* (the latest reboot of the perennial property, directed by Jonathan Liebesman) share a very similar audiovisual aesthetic. *Transformers* is the more overtly science fiction film of the two—it even opens with the familiar *Star Wars* trope depicting a spaceship passing overhead towards a planet below, complete with the sound of the ship passing in the overhead channels. Unlike James Gunn, who has affection for the old

technology of the cassette player in *Guardians of the Galaxy*, Bay is scornful of outdated media technology. In an early scene, the protagonists visit an derelict old cinema to purchase disused film stock and projection equipment being sold for scrap. Bay's heroes have no interest in the cinematic technology for sentimental reasons—the machines are presumably taken apart in order to create new inventions in the spirit of American entrepreneurialism. Bay has no patience for outdated modes of technology, and accordingly the audiovisual aesthetics of *Transformers* and *Teenage Mutant Ninja Turtles* fully embrace the affordances of the Atmos system. In both movies, shots are framed from low angles, and CGI creatures and robots freely move around and above the frame to highlight the possibilities of the expanded speaker arrays through the theater. These movies take full advantage of the design of Atmos in a spectacular manner that recalls flashy uses of 3D visual effects that constantly move objects towards the audience in order to bring attention to the depth of the image. Unlike *Guardians of the Galaxy*'s ambivalent relationship towards Atmos, Bay's movies seem to construct action set pieces with the format in mind from the start with an emphasis on the capabilities of the overhead speakers.

Moving from Bay's scifi-action to the straightforward action, *The Expendables 3*, directed by Patrick Hughes and featuring a cast of 1980s action stalwarts, is essentially an extended homage to the history of the genre. The movie's debts to previous action films are considerably more overt even than *Guardians of the Galaxy*'s relationship to science fiction. Throughout the film, Stallone delivers innumerable speeches centered about his character's increasing obsolescence in the current marketplace of high-priced mercenaries. The dialogue carries an very thinly veiled subtext about the aging actor's dissatisfaction with waning fortunes in show business. With this sense of nostalgia permeating the movie, it's no surprise that the audiovisual aesthetics are not particularly formally inventive. There are a few obvious

demonstrations of the capabilities of the overhead speakers when a train and then a helicopter pass overhead in the opening minutes of the movie. But after this obligatory display, the movie unsurprisingly reverts to the usual 5.1 digital surround style. Only the occasional smattering of gunfire and explosions remind the auditor that the soundtrack features a post-channel, object-based mix.

Finally, the soundtrack for dance movie *Step Up: All In* (directed by Trishe Sie) proves to be the most incongruous fit with the strictures of Atmos. The *Step Up* series is closely aligned with the tradition of the Hollywood musical, the classical film genre that most commonly flouted realism conventions. The characters in *Step Up: All In* never burst into song, but they do often break into fantastical dance numbers that strain credulity. In these dance scenes, the music maintains a complicated relationship to the diegesis. The dancers must be able to hear the music within the diegetic world, but for issues of aural legibility it cannot be specifically placed within the diegesis—something that Atmos was specifically designed to do. In *Step Up: All In*, supervising sound editor Odin Benitez takes an unusual tack. For most of the dance sequences, the sound emanates from the from the front speakers at all times, even as the camera cuts to different points of view, and characters are heard from the corresponding points of audition. The consistent placement of the sound object within the theater's speaker arrangement irregardless of the diegesis again highlights how Atmos is not well-suited for ambiguous relationships between sound and image.

CONCLUSION: CULTURAL RESONANCES

The ultimate success of Atmos remains an open question. After nearly three years, it is off to a sluggish start compared to previous Dolby formats. Four years after the introduction of the company's first format, Dolby Stereo, 2,000 theaters in the US were equipped with the system.[124] By 1995, three years after the introduction of Dolby Digital, 800 theaters in the US had upgraded, although the format would expand exponentially in the following years.[125] Four years after the introduction of Dolby EX, the first format for DCP, over 4,000 theaters had upgraded.[126] But at this point, there are only around 300 Atmos screens in the United States.[127] This is due in part to the hefty $30,000 cost per screen to make the upgrade, coupled with the failure of Atmos to be associated as the definitive theatrical format for discerning cinephiles in the way similar to how IMAX has caught on thanks to the support of Christopher Nolan. But regardless of the format's long-term success or failure, the emergence of Atmos as the first format designed for digital cinema projection says something about the particular cultural moment that produced it.

First and foremost, Atmos is symptomatic of a particular ethos of the current moment—typified by the deification of big data—that reflexively privileges machinic objectivity over subjective and/or affective responses. With this ideology, it is clear that computers and other tools can perceive more information more consistently than human beings could ever hope to. The distrust of unquantifiable, subjective understandings or interpretations is apparent in the

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[124] Sergi, *The Dolby Era*, 30.
[125] Natale, Richard. "Company Town: Three Companies Wage Battle for the Hearts--and Ears--of U.S. Moviegoers." *Los Angeles Times*, September 5, 1995.
[126] "Dolby Atmos Next-Generation Audio for Cinema." Dolby Laboratories, 2014.
[127] "100 Dolby Atmos Screens with Cinema 21." *Dolby Investor Relations*, December 9, 2014.

mystical celebration of the powers of big data and the growth of certain strains of objective research in the digital humanities currently in vogue within the academy. Atmos is a clear manifestation of this cultural moment. It is a format that bases it conceptions of fidelity according to objective machine perception and turns over the formerly subjective task of mixing and panning to an algorithm that consistently places sounds in the correct positions. With Atmos, the affective possibilities of sound cede ground to the constraints of machinic objectivity.[128] Additionally, Atmos marks another turning point in the production history of film sound as sound designers are forced into the role of postindustrial producer, according to Alan Liu's definition of the concept.[129] Atmos severs the relationship between form and content, forcing filmmakers to create soundtracks according to the strict spatial regimes of the object-based format.

Areas for Further Research

I should acknowledge that the ideal viewer I have suggested filmmakers imagining and the customer Dolby is courting occupy a particular socioeconomic position. Barbara Klinger has written convincingly about how home theater companies conflate spending power with connoisseurship in advertising high-end audiovisual technology.[130] It should go without saying that a consumer must have a substantial disposable income in order to purchase Atmos equipment—even admission to an Atmos-equipped theater usually means inflated 3D ticket prices. However, home theater audio enthusiasts are not only economically privileged, they are also overwhelmingly male. Delving into the subculture of audiophiles interested in high-end

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[128] Vessa, Brian, and Kurt Graffy. *Frequency Response Analysis of Theaters and Dubbing Stages*.
[129] Liu, Alan. "Transcendental Data: Toward a Cultural History and Aesthetics of the New Encoded Discourse."
[130] Klinger, *Beyond the Multiplex*, 38-9.

audio equipment reveals a community of predominately older, middle-class white men enthusing over the latest technological advancement that Dolby has released. Websites like twit.com and podcasts like Scott Wilkinson's *Home Theater Geeks* are startlingly insular spaces in which middle-aged men complain about their wives' lack of understanding of audio technology.[131] These websites are primary places were Dolby disseminates information about their new products—even Ioan Allen, longtime Vice President of Dolby Laboratories has been interviewed on Wilkinson's podcasts. The blatant misogyny and unexamined privilege of the audiophile community is seemingly authorized by the members' vaguely scientific expertise in matters of audio technology. Questions of objectivity and scientific accuracy are often engaged, revealing a general preference for accuracy and high fidelity over creative interpretations in sonic representation. It's possible to analyze sound aesthetics without paying attention to the specific moments of reception of new texts and technologies, but the success of Atmos will largely depend on its adoption and advocacy by this specialist community of audiophiles. So I believe it's important to recognize their particular ideological grounding, and this interest in objectivity and realism shapes and is shaped by Dolby's marketing. Furthermore, as Keir Keightley has demonstrated in his study of post-war hi-fi culture, the very concept of sonic "immersion" has fraught roots in misogynist discourse against supposedly feminized domestic space.[132]

Finally, my analysis has centered only Hollywood films, but Atmos is also increasingly popular internationally, particularly in a few Asian markets. Due to issues of access, it was

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[131] See Scott Wilkinson, *Home Theater Geeks*. http://twit.tv/show/home-theater-geeks
[132] Keightley, Keir. "'Turn It down!' She Shrieked: Gender, Domestic Space, and High Fidelity, 1948-59." *Popular Music* 15, no. 2 (May 1, 1996): 149–77.

impossible for me to hear other film culture's approaches to Atmos, but understanding other audiovisual aesthetic approaches would be a boon to scholarship of immersive sound.

BIBLIOGRAPHY

"100 Dolby Atmos Screens with Cinema 21." *Dolby Investor Relations*, December 9, 2014. http://investor.dolby.com/releasedetail.cfm?ReleaseID=886677.

"2014 Domestic Grosses." *Box Office Mojo*. Accessed February 16, 2015. http://www.boxofficemojo.com/yearly/chart/?yr=2014.

"2014 Worldwide Grosses." *Box Office Mojo*. Accessed February 16, 2015. http://www.boxofficemojo.com/yearly/chart/?view2=worldwide&yr=2014.

Alimurung, Gendy. "Movie Studios Are Forcing Hollywood to Abandon 35mm Film. But the Consequences of Going Digital Are Vast, and Troubling." *LA Weekly*, April 12, 2012. http://www.laweekly.com/film/movie-studios-are-forcing-hollywood-to-abandon-35mm-film-but-the-consequences-of-going-digital-are-vast-and-troubling-2174582.

Altman, Rick. "Introduction: Four and a Half Film Fallacies." In *Sound Theory, Sound Practice*, edited by Rick Altman, 35–45. New York: Routledge, 1992.

———. *Silent Film Sound*. New York: Columbia UP, 2004.

———. "Sound Space." In *Sound Theory, Sound Practice*, edited by Rick Altman, 46–64. New York: Routledge, 1992.

———. , ed. *Sound Theory, Sound Practice*. New York: Routledge, 1992.

Andrews, Mark, Brenda Chapman, and Steve Purcell. *Brave*. Animation, Adventure, Comedy, 2012.

Attali, Jacques. *Noise: The Political Economy of Music*. Translated by Brian Massumi. Minneapolis: U of Minnesota P, 1985.

"Auro 11.1 - Barco's Immersive Cinema Sound Format," 2015. http://www.barco.com/en/Auro11-1.

Barthes, Roland. *Mythologies*. Translated by Annette Lavers and Richard Howard. New York: Hill and Wang, 2013.

Baudrillard, Jean. *Simulacra and simulation*. Translated by Sheila Faria Glaser. Ann Arbor: U of Michigan P, 1994.

Bay, Michael. *Transformers: Age of Extinction*. Blu-ray. Paramount, 2014.

Bazin, Andre. *What Is Cinema?*. Translated by Hugh Gray. Berkeley: University of California Press, 1967.

Beck, Jay Shields. “A Quiet Revolution: Changes in American Film Sound Practices, 1967-1979.” Ph.D. diss., University of Iowa, 2003.

Belton, John. “1950s Magnetic Sound: A Frozen Revolution.” In *Sound Theory, Sound Practice*, edited by Rick Altman, 154–67. New York: Routledge, 1992.

———. “Digital Cinema: A False Revolution.” *October* 100 (April 1, 2002): 99–114.

———. “Technology and Aesthetics of Film Sound.” In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, 63–72. New York: Columbia UP, 1985.

———. *Widescreen Cinema*. Cambridge, MA: Harvard UP, 1992.

Benjamin, Walter. “The Work of Art in the Age of Its TechnologicoI Reproducbility.” In *Walter Benjamin: Selected Writings, Volume 4: 1938-1940*, edited by Marcus Paul Bullock and Howard Eiland, translated by Edmund Jephcott, 251–71. Cambridge, Mass.: Belknap Press, 2004.

Boeddeker, Steve. *Dolby Presents: Silent, a Short Film*. Moonbot Studios, 2014. https://vimeo.com/86542938.

Bolter, J. David, and Richard A Grusin. *Remediation: Understanding New Media*. Cambridge, MA: MIT Press, 2000.

Bordwell, David. *Pandora’s Digital Box: Films, Files, and the Future of Movies*. Madison, WI: Irvington Way Institute P, 2013.

———. *The Way Hollywood Tells It: Story and Style in Modern Movies*. Berkeley: U of California P, 2006.

Child, Ben. “Ant-Man: Kevin Feige Rejects Marvel Criticism over Edgar Wright Exit.” *The Guardian*, July 16, 2014. http://www.theguardian.com/film/2014/jul/16/ant-man-kevin-feige-rejects-marvel-criticism-over-edgar-wright-exit.

Chion, Michel. *Audio-Vision: Sound on Screen*. Edited and translated by Claudia Gorbman. New York: Columbia UP, 1994.

———. *Film, A Sound Art*. Translated by Claudia Gorbman. New York: Columbia UP, 2009.

———. “Quiet Revolution... And Rigid Stagnation.” Translated by Ben Brewster. *October* 58 (October 1, 1991): 69–80. doi:10.2307/778798.

“Cinema & Entertainment: Solutions for Cinemas.” *IOSONO: The Future of Spatial Audio*. Accessed February 24, 2015. http://www.iosono-sound.com/events-entertainment/solutions-for/cinemas/.

"Cinemascope: What It Is: How It Works." *American Cinematographer*, March 1953. http://www.widescreenmuseum.com/widescreen/cscope-ac.htm.

Cousins, Mark. "Walter Murch." In *Projections: Film-Makers on Film-Making*, edited by John Boorman and Walter Donohue, 149–62. London: Faber, 1996.

Crowther, Bosley. "Sound and (or) Fury: Stereophonic System Is Debated in Hollywood." *The New York Times*, January 31, 1954. http://query.nytimes.com/gst/abstract.html?res=9903E6DF143AE53BBC4950DFB766838F649EDE.

Disentfrey, Eric. "Dolby Atmos: What You Hear | Antenna." *Antenna*, April 11, 2013. http://blog.commarts.wisc.edu/2013/04/11/dolby-atmos-what-you-hear/.

Doane, Mary Ann. "Ideology and the Practice of Sound Editing and Mixing." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, 54–62. New York: Columbia UP, 1985.

Dolby Atmos: "Amaze" Cinema Trailer, 2014. http://www.youtube.com/watch?v=kvAfmYNtugQ&feature=youtube_gdata_player.

Dolby Atmos: "Conductor" Cinema Trailer. Youtube, 2014. https://www.youtube.com/watch?v=3TOlN9dLpi8&feature=youtube_gdata_player.

"Dolby Atmos Home Theater Products." *Dolby.com*. Accessed March 5, 2015. /us/en/technologies/dolby-atmos/home-theater-products.html.

Dolby Atmos: "Leaf" Cinema Trailer. Youtube, 2014. https://www.youtube.com/watch?v=qJA2U-YMvkk&feature=youtube_gdata_player.

"Dolby Atmos Movies on Blu-Ray." *Dolby.com*. Accessed February 16, 2015. /us/en/experience/dolby-atmos/bluray-and-streaming.html.

"Dolby Atmos Multidimensional Sound." *Dolby.com*. Accessed September 26, 2014. http://www.dolby.com/us/en/technologies/dolby-atmos.html.

"Dolby Atmos Next-Generation Audio for Cinema." Dolby Laboratories, 2014. http://www.dolby.com/us/en/professional/cinema/products/dolby-atmos-next-generation-audio-for-cinema-white-paper.pdf.

Dolby Atmos: "Unfold" Cinema Trailer -, 2014. https://www.youtube.com/watch?v=nBITovLH6So&feature=youtube_gdata_player.

"Dolby Digital Surround EX." *Dolby.com*. Accessed February 24, 2015. /us/en/technologies/dolby-digital-surround-ex.html.

Dolby Digital Trailer - Train (original), 2009. https://www.youtube.com/watch?v=suEkdOo30NY&feature=youtube_gdata_player.

"Dolby Surround 7.1." *Dolby.com*. Accessed February 24, 2015. /us/en/technologies/dolby-surround-7-1.html.

Donner, Richard. *Superman - The Movie*. DVD. Warner Home Video, 2013.

Eisenstein, Sergei, Vsevolod Pudovkin, and G.V. Alexandrov. "A Statement." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, translated by Jay Leyda, 83–85. New York: Columbia UP, 1985.

Ferguson, Norman, James Algar, Samuel Armstrong, Ford Beebe Jr, Jim Handley, T. Hee, Wilfred Jackson, et al. *Fantasia*. Animation, Family, Fantasy, 1940.

Ford, John. *Stagecoach*. Criterion, 2010.

Forlenza, Jeff, and Terri Stone, eds. *Sound for Picture: An Inside Look at Audio Production for Film and Television*. Emeryville, CA: Hal Leonard Corp, 1993.

Fuchs, Andreas. "Soundsational! Dolby Atmos Delivers Exciting New Audio Platform." *Film Journal*, April 20, 2012. http://www.filmjournal.com/content/soundsational-dolby-atmos-delivers-exciting-new-audio-platform.

Giardina, Carolyn. "Christopher Nolan Breaks Silence on 'Interstellar' Sound." *The Hollywood Reporter*, November 5, 2014. http://www.hollywoodreporter.com/behind-screen/christopher-nolan-breaks-silence-interstellar-749465.

———. "Dolby's New 'Atmos' System Rockets Movie Sound to the Next Level." *The Hollywood Reporter*, July 12, 2012. http://www.hollywoodreporter.com/news/dolby-atmos-brave-348083.

———. "LAFF: Dolby Atmos Debuts at Disney/Pixar's 'Brave' Premiere." *The Hollywood Reporter*, June 9, 2012. http://www.hollywoodreporter.com/news/dolby-atmos-brave-premiere-pixar-laff-339193.

———. "'Oblivion,' 'The Croods,' 'Monsters University,' 'Iron Man 3' Will Get Atmos Treatment - Hollywood Reporter." *The Hollywood Reporter*, February 14, 2013. http://www.hollywoodreporter.com/news/oblivion-croods-monsters-university-iron-421414.

Gitelman, Lisa. *Always Already New: Media, History and the Data of Culture*. Cambridge, Mass.; London: MIT, 2008.

Godard, Jean-Luc. *Goodbye to Language*. Kino Lorber, 2014.

Gomery, Douglas. "Economic Struggle and Hollywood Imperialism: Europe Converts to Sound." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, 25–36. New York: Columbia UP, 1985.

———. "The Coming of Sound: Technological Change in the American Film Industry." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, 5–24. New York: Columbia UP, 1985.

Gunn, James. *Guardians of the Galaxy*. Dolby Digital theatrical mix, Action, Adventure, Sci-Fi. Walt Disney, 2014.

———. *Guardians of the Galaxy*. Dolby Atmos theatrical mix, Action, Adventure, Sci-Fi. Walt Disney, 2014.

———. *Slither*. DVD. Universal Studios, 2006.

———. *Super*. DVD. MPI Home Video, 2011.

Handzo, Stephen. "A Narrative Glossary of Film Sound Technology." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, 383–426. New York: Columbia UP, 1985.

"Home Theater Speaker Guide." Dolby Laboratories, 2010. http://www.dolby.com/uploadedFiles/Assets/US/Doc/Consumer/Dolby-Home-Theatre-Speaker-Guide-7.1-6-8.pdf.

Horkheimer, Max, and Theodor W. Adorno. *Dialectic of Enlightenment*. Translated by John Cumming. Continuum, 1972.

Hughes, Patrick. *The Expendables 3*. Blu-ray. Lionsgate, 2014.

Hull, Joseph. "Surround Sound Past, Present, and Future." Dolby Laboratories, 1999. http://pacificav.com/library/Dolby%20Surround%20History.pdf.

Jameson, Frederic. "Postmodernism and Consumer Society." In *The Norton Anthology of Theory and Criticism*, edited by Vincent B. Leitch, First Ed., 1960–74. New York: W. W. Norton, 2001.

Jenkins, Henry. *Convergence Culture: Where Old and New Media Collide*. Revised edition. New York: NYU Press, 2008.

Keightley, Keir. "'Turn It down!' She Shrieked: Gender, Domestic Space, and High Fidelity, 1948-59." *Popular Music* 15, no. 2 (May 1, 1996): 149–77.

Kellogg, Edward W. "Final Installment: History of Sound Motion Pictures." *SMPTE Journal* 64, no. 8 (1955): 422–37.

———. "Second Installment: History of Sound Motion Pictures." *SMPTE Journal* 64, no. 7 (1955): 356–74.

Kenny, Tom. "Introduction." In *Sound for Picture: An Inside Look at Audio Production for Film and Television*, edited by Jeff Forlenza and Terri Stone, v – vi. Emeryville, CA: Hal Leonard Corp, 1993.

Kerins, Mark. *Beyond Dolby (Stereo): Cinema in the Digital Sound Age*. Bloomington: Indiana UP, 2010.

Kilcoyne, Sean. "You Shouldn't Have Been That Sentimental." *Journal of Information Ethics* 19, no. 1 (April 1, 2010): 57–73. doi:10.3172/JIE.19.1.57.

Klinger, Barbara. *Beyond the Multiplex: Cinema, New Technologies, and the Home*. Berkeley: U of California P, 2006.

Koster, Henry. *The Robe*. DVD. Fox, 2009.

Lastra, James. *Sound Technology and the American Cinema: Perception, Representation, Modernity*. New York: Columbia UP, 2000.

Liebesman, Jonathan. *Teenage Mutant Ninja Turtles*. Blu-ray. Paramount, 2014.

Lipscomb, Scott D., and Roger A. Kendall. "Perceptual Judgement of the Relationship between Musical and Visual Components in Film." *Psychomusicology: Music, Mind & Brain* 13, no. 1 (1994): 60–98.

Liu, Alan. "Transcendental Data: Toward a Cultural History and Aesthetics of the New Encoded Discourse." *Critical Inquiry* 31, no. 1 (2004): 49–84.

Lucas, George. "Movies Are an Illusion." *Premiere* 60 (February 1999).

———. *Star Wars: Episode I - The Phantom Menace*. Action, Adventure, Fantasy, 1999.

———. *Star Wars Trilogy*. DVD. 20th Century Fox, 2004.

Malick, Terrence. *Days of Heaven*. DVD. Criterion, 2007.

Metz, Christian. "Aural Objets." In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, translated by Georgia Gurrieri, 154–61. New York: Columbia UP, 1985.

Morrison, Geoffrey. “Dolby Atmos at Home: Ears on.” *CNET*, Agust 2014. http://www.cnet.com/news/dolby-atmos-at-home-ears-on/.

Natale, Richard. “Company Town: Three Companies Wage Battle for the Hearts--and Ears--of U.S. Moviegoers.” *Los Angeles Times*, September 5, 1995. http://articles.latimes.com/1995-09-05/business/fi-42479_1_digital-sound.

Negulesco, Jean. *How To Marry A Millionaire*. DVD. Twentieth Century Fox, 2001.

Pinch, Trevor. “Technology and Institutions: Living in a Material World.” *Theory and Society* 37, no. 5 (July 10, 2008): 461–83. doi:10.1007/s11186-008-9069-x.

Pudovkin, Vsevolod. “Asynchronism as a Principle of Sound FIlm.” In *Film Sound: Theory and Practice*, edited by Elisabeth Weis and John Belton, translated by Marie Seton and Ivor Montagu, 86–91. New York: Columbia UP, 1985.

Rodowick, David Norman. *The Virtual Life of Film*. Cambridge: Harvard UP, 2007.

Scott, Ridley. *Blade Runner*. DVD. Warner Home Video, 2013.

Sergi, Gianluca. “Knocking at the Door of Cinematic Artifice: Dolby Atmos, Challenges and Opportunities.” *The New Soundtrack* 3, no. 2 (August 28, 2013): 107–21. doi:10.3366/sound.2013.0041.

———. *The Dolby Era: Film Sound in Contemporary Hollywood*. New York: Manchester UP, 2004.

Shaviro, Steven. *Post Cinematic Affect*. Hants, UK: Zero Books, 2010.

“Shoot Film, Don’t Kill It.” *Savefilm.org*. Accessed February 24, 2015. http://www.savefilm.org/.

Sie, Trish. *Step Up All In*. Blu-ray. Lionsgate, 2014.

Smith, Jeff. “Atmos, All around: A Guest Post by Jeff Smith.” *Observations on Film Art*, May 1, 2013. http://www.davidbordwell.net/blog/2013/05/01/atmos-all-around-a-guest-post-by-jeff-smith/.

Sobchack, Vivian. “When the Ear Dreams: Dolby Digital and the Imagination of Sound.” *Film Quarterly* 58, no. 4 (June 1, 2005): 2–15. doi:10.1525/fq.2005.58.4.2.

Sobchack, Vivian Carol. *Screening Space: The American Science Fiction Film*. New York: Ungar, 1987.

Sontag, Susan. “The Imagination of Disaster.” *Commentary*, October 1965.

Spielberg, Steven. *Saving Private Ryan*. DVD. Dreamworks Video, 1999.

Sterne, Jonathan. *MP3: The Meaning of a Format*. Durham: Duke UP, 2012.

———. “Sonic Imaginations.” In *The Sound Studies Reader*, edited by Jonathan Sterne, 1–21. New York: Routledge, 2012.

———. *The Audible Past: Cultural Origins of Sound Reproduction*. Durham: Duke UP, 2003.

———. , ed. *The Sound Studies Reader*. New York: Routledge, 2012.

Théberge, Paul, Kyle Devine, and Tom Everrett, eds. *Living Stereo: Histories and Cultures of Multichannel Sound*. New York: Bloomsbury Academic, 2015.

Unkrich, Lee. *Toy Story 3*. Animation, Adventure, Comedy, 2010.

Vessa, Brian, and Kurt Graffy. *Frequency Response Analysis of Theaters and Dubbing Stages*. White Plains, NY: Society of Motion Picture and Television Engineers (SMPTE), October 1, 2014. https://www.smpte.org/standards/reports.

Weis, Elisabeth, and John Belton, eds. *Film Sound: Theory and Practice*. New York: Columbia UP, 1985.

Whittington, William. *Sound Design and Science Fiction*. Austin, TX: U of Texas P, 2007.

Wilkinson, Scott. *40 Years of Dolby*. Mp3. Home Theater Geeks, n.d. http://twit.tv/show/home-theater-geeks/168.

———. *Auro-3D Immersive Audio*. Mp3. Home Theater Geeks. Accessed September 26, 2014. http://twit.tv/show/home-theater-geeks/230.

———. *Cinema Sound: Now and Beyond*. Mp3. Home Theater Geeks. Accessed September 26, 2014. http://twit.tv/show/home-theater-geeks/230.

———. *Dolby Atmos Questions Answered*. Mp3. Home Theater Geeks. Accessed September 29, 2014. http://twit.tv/show/home-theater-geeks/222.

———. *The Sorry State of Cinema Sound*. Mp3. Home Theater Geeks. Accessed September 26, 2014. http://twit.tv/show/home-theater-geeks/229.

Wright, Benjamin. “Atmos Now: Dolby Laboratories, Mixing Ideology and Hollywood Sound Production.” In *Living Stereo: Histories and Cultures of Multichannel Sound*, edited by Paul Théberge, Kyle Devine, and Tom Everrett, 227–46. New York: Bloomsbury Academic, 2015.